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YM310 A19 CAT1 Module Hardware Manual

V1.1



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Chapter 1 Introduction

This document is the hardware interface manual of the wireless solution product YM310 A19 LGA package CAT1 module, aiming to describe the hardware composition and functional features, application interface definitions and usage instructions, electrical performance and mechanical characteristics, etc. of this module solution product. Combined with this document and other application documents, users can quickly use this module to design wireless application solutions.

Chapter 2. Module Overview

2.1 Module Introduction

YM310 A19 is a small-sized, high-performance and ultra-low power consumption LTE Cat.1 wireless communication module developed based on the ASR1605 platform. The module supports LTE-FDD and LTE-TDD network data connection communication protocols, and supports LTE 3GPP R13 Cat1, with a maximum download rate of 10Mbps and an upload rate of 5Mbps. Support WiFi SCAN function.

YM310 A19 is designed as a chip module, using LGA surface mount packaging, with a total of 109 pins. The module size is only (17.7 ± 0.1) mm * (15.8 ± 0.1) mm * (2.3 ± 0.15) mm. YM310 A19 is small in size and has large applications. It has the same size as mainstream GSM and NB-IOT modules and can meet almost all M2M application needs.

The YM310 A19 module can be applied in the following occasions:

- ✧ Security system
- ✧ Wireless POS machine
- ✧ Wireless advertising, multimedia
- ✧ Remote monitoring
- ✧ Smart meter reading
- ✧ Mobile broadband
- ✧ Automated industry
- ✧ Other wireless terminals, etc

2.2 Module Characteristics

Table 2-1 List of Module Models

Model	Band	Chip（RAM+Flash）
YM310 A19AEUCX	LTE:B1/3/7/8/20/28/38/40/41	ASR1605C（4MB+2MB）
YM310 A19AEVCX	LTE:B1/3/5/7/8/20/28/38/40/41	
YM310 A19ANACX	LTE:B2/4/5/12/13/17/66/38/41	
YM310 A19ANECX	LTE:B2/4/5/12/13/17/66/71/38/41	

YM310 A19AJPCX	LTE:B1/3/8/18/19/26/28/38/40/41	
YM310 A19ASACX	LTE:B2/4/5/7/28/66/38/40/41	
YM310 A19AEUSX	LTE:B1/3/7/8/20/28/38/40/41	ASR1605S（4MB+4MB）
YM310 A19AEVSX	LTE:B1/3/5/7/8/20/28/38/40/41	
YM310 A19ANASX	LTE:B2/4/5/12/13/17/66/38/41	
YM310 A19ANESX	LTE:B2/4/5/12/13/17/66/71/38/41	
YM310 A19AJPSX	LTE:B1/3/8/18/19/26/28/38/40/41	
YM310 A19ASASX	LTE:B2/4/5/7/28/66/38/40/41	

Table 2-2 Key Characteristics

Features		**Description**
Physical Property		(17.7±0.1)mm * (15.8±0.1)mm * (2.3±0.15)mm
Mode of Fixation		LGA packaging, fixed with surface mount
Working Voltage		3.3V - 4.2V Typical Voltage: 3.7V
Energy Saving Current		The dormant current is less than 1.1mA
Application Interface	USIM Interface	✧ Supports two sets of USIM cards, among which USIM1 supports 3.0V and 1.8V. USIM1 supports hot-swappable function ✧ The USIM2 only supports 1.8V USIM cards
	USB Interface	✧ Complies with the USB2.0 specification (only supports slave mode), with a maximum data transfer rate of 480Mbps ✧ Used for AT commands, data transmission, software debugging and software upgrades ✧ The USB driver supports Windows/Linux/Android, etc
	UART Interface	Main serial port (4 wires) : ✧ Used for AT commands and data transmission ✧ Support RTS and CTS hardware flow control ✧ The maximum baud rate supported is 3.6Mbps, and the default is 115200bps Auxiliary serial port (2 wires) : ✧ Used for communicating with peripheral devices ✧ The default baud rate is 115200bps

r f a c e		Debug serial port (2 wires) : ✧ It is used for debugging information output and printing module logs ✧ The default baud rate is 115200bps
	I2C Interface	✧ It conforms to the I2C bus protocol ✧ The high-speed mode can support a maximum rate of 3.4Mbps
	PCM Interface	✧ Support a set of PCM digital voice to achieve audio output with external DAC or CODEC devices ✧ The PCM_DOUT&DIN pin function software can be flexibly configured
	ADC Interface	✧ Supports 2 channels of 12-bit sampling ADC ✧ The voltage input range is 0 to 1.2V
	Status Indication	✧ NET_STATUS indicates the network operation status ✧ The STATUS module indicates the running status
	BOOT	✧ Support 1 forced download interface
LTE Transmission Power		✧ LTE: Class 3(23dBm±2dB)
LTE Characteristics		✧ LTE: DL 10Mbps, UL 5Mbps@20M ✧ Support 3GPP Rel-13 Cat1 bis FDD and TDD ✧ Supports 1.4/3/5/10/15/20 MHZ RF bandwidth
Short Message (SMS)		✧ Text and PDU mode ✧ Point-to-point text message sending and receiving ✧ Short message community broadcast ✧ Short message storage: Stored in the USIM card and ME, with the default being in ME
WiFi Scan		✧ Support WIFI Scan positioning (share main antenna, time division multiplexing)
AT Command		✧ Support the standard AT instruction set (Hayes 3GPP TS 27.007 and 27.005) ✧ Specifically query the AT instruction set
Network Protocol		✧ Support TCP/UDP/PPP/HTTP/NITZ/CMUX/ RNDIS/NTP/HTTPS/PING Agreement

Antenna Interface	✧ ANT_MAIN *1, Characteristic Impedance 50Ω
Virtual Network Card	✧ Support USB virtual network card
Temperature Range	✧ Normal operating temperature: -30℃ to + 75℃ ✧ maximum operating temperature range is from -40℃ to + 85℃ ✧ Storage temperature: -40℃ to + 90℃

NOTE

✧ When the temperature is within the range of -40 ℃ to -30 ℃ or +75 ℃ to +85 ℃, individual RF indicators of the module may slightly exceed the 3GPP standard range. The module can still maintain a normal working state, and the radio frequency spectrum and network are basically not affected. When the temperature returns to the normal operating temperature range, all the indicators of the module can still meet the requirements of the 3GPP specification.

2.3 Module Functions

YM310 A19 module mainly contains the following circuit units:

- ✧ LTE Radio frequency processing unit
- ✧ Power management unit
- ✧ Module interface unit
- ✧ Peripheral interface unit

The functional block diagram of the YM310 A19 module is shown as follows:

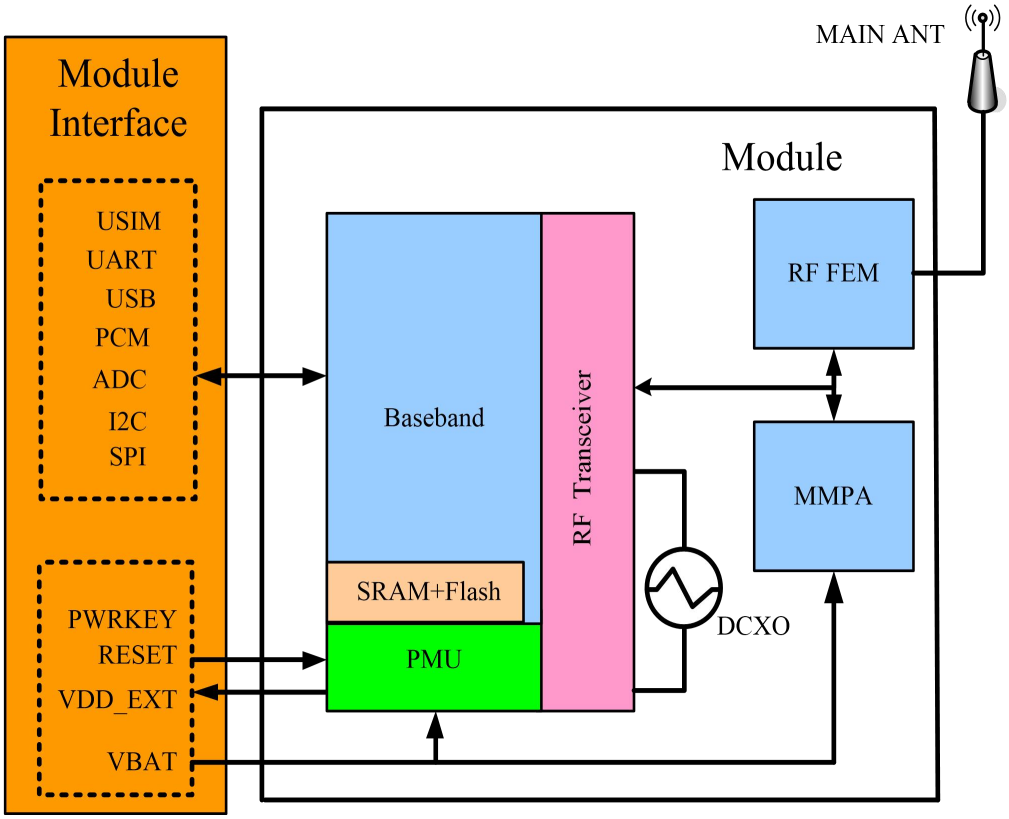


Figure 2-1 Functional block diagram of YM310 A19 module

Chapter 3 Interface Application Description

3.1 Overview of This Chapter

This chapter mainly describes the interface definition and application of this module. It includes the following parts:

- ✧ Module Pin Distribution Diagram
- ✧ Pin Definition
- ✧ Power Interface
- ✧ USB Interface
- ✧ USIM Interface
- ✧ UART Interface
- ✧ ADC Interface
- ✧ PCM Interface
- ✧ Status Indication Interface
- ✧ Radio Frequency Antenna Interface

3.2 Module Interface

3.2.1 Module Pin Distribution Diagram

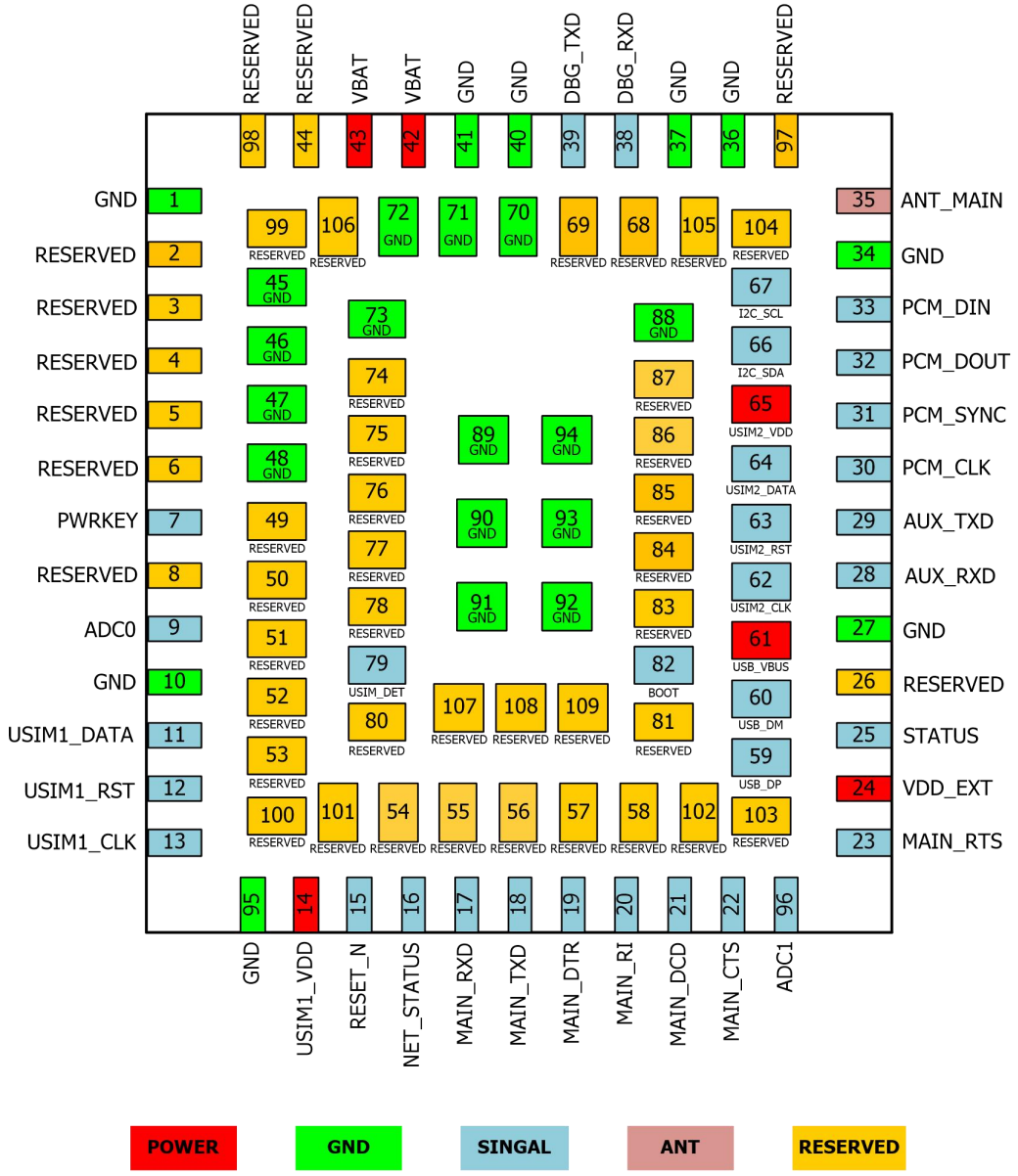


Figure 3-1 Pin Distribution Diagram of YM310 A19 module (TOP Perspective)

NOTE

- ✧ All RESERVED pins need to be left floating.
- ✧ Module 82 pins are not allowed to be pulled down before the module is successfully powered on.
- ✧ To be compatible with the CLM920_HV9&LV9 module, pins PIN32 and PIN33 can be configured as PCM_DIN and PCM_DOUT respectively through software.

3.2.2 Pin Definition

The YM310 A19 module is an LGA interface module, and its pin definitions are shown in the following table:

Table 3-1 Pin Definitions

Serial Number	Pin Name	Serial Number	Pin Name
1	GND	2	RESERVED
3	RESERVED	4	RESERVED
5	RESERVED	6	RESERVED
7	PWRKEY	8	RESERVED
9	ADC0	10	GND
11	USIM1_DATA	12	USIM1_RST
13	USIM1_CLK	14	USIM1_VDD
15	RESET_N	16	NET_STATUS
17	MAIN_RXD	18	MAIN_TXD
19	MAIN_DTR	20	MAIN_RI
21	MAIN_DCD	22	MAIN_CTS
23	MAIN_RTS	24	VDD_EXT
25	STATUS	26	RESERVED
27	GND	28	AUX_RXD
29	AUX_TXD	30	PCM_CLK
31	PCM_SYNC	32	PCM_DOUT
33	PCM_DIN	34	GND
35	ANT_MAIN	36	GND
37	GND	38	DBG_RXD
39	DBG_TXD	40	GND
41	GND	42	VBAT
43	VBAT	44	RESERVED
45	GND	46	GND
47	GND	48	GND
49	RESERVED	50	RESERVED
51	RESERVED	52	RESERVED

53	RESERVED	54	RESERVED
55	RESERVED	56	RESERVED
57	RESERVED	58	RESERVED
59	USB_DP	60	USB_DM
61	USB_VBUS	62	USIM2_CLK
63	USIM2_RST	64	USIM2_DATA
65	USIM2_VDD	66	I2C_SDA
67	I2C_SCL	68	RESERVED
69	RESERVED	70	GND
71	GND	72	GND
73	GND	74	RESERVED
75	RESERVED	76	RESERVED
77	RESERVED	78	RESERVED
79	USIM1_DET	80	RESERVED
81	RESERVED	82	BOOT
83	RESERVED	84	RESERVED
85	RESERVED	86	RESERVED
87	RESERVED	88	GND
89	GND	90	GND
91	GND	92	GND
93	GND	94	GND
95	GND	96	ADC1
97	RESERVED	98	RESERVED
99	RESERVED	100	RESERVED
101	RESERVED	102	RESERVED
103	RESERVED	104	RESERVED
105	RESERVED	106	RESERVED
107	RESERVED	108	RESERVED
109	RESERVED		

Table 3-2 Definitions of IO Parameters

Symbolic Mark	**Description**

IO	Two-Way Input and Output
PI	Power Input
PO	Power Output
AI	Analog Input
AO	Analog Output
AIO	Analog input/output
DI	Digital Input
DO	Digital Output
OD	Open Drain

Table 3-3 Pin Description

Power Supply				
Pin Number	**Pin Definition**	**IO**	**Function Description**	**Remarks**
14	USIM1_VDD	PO	USIM1 power supply	The module automatically recognizes 1.8V or 3.0V SIM cards
65	USIM2_VDD	PO	USIM2 power supply	Only 1.8V SIM cards are supported
24	VDD_EXT	PO	1.8V voltage output	It can provide pull-up for external GPIOs (Imax<50mA)
42，43	VBAT	PI	Module main power supply	The power supply needs to provide a minimum current-carrying capacity of 1.5A
61	USB_VBUS	PI	USB insertion detection	Range of 3~5.25V
1，10，27，34，36-37，40-41，45-48，70-73，88-95			GND	
Module Status Display Interface				

Pin Number	Pin Definition	IO	Function Description	Remarks
16	NET_STATUS	DO	Module network status indication	1.8V level. If not in use, it remains suspended
25	STATUS	DO	Module operation status indication	1.8V level. If not in use, it remains suspended
Debug Serial Port				
Pin Number	**Pin Definition**	**IO**	**Function Description**	**Remarks**
38	DBG_RXD	DI	Debug the serial port data reception	1.8V level. If not in use, it remains suspended
39	DBG_TXD	DO	Debug the serial port data transmission	1.8V level. If not in use, it remains suspended
SIM Interface				
Pin Number	**Pin Definition**	**IO**	**Function Description**	**Remarks**
11	USIM1_DATA	IO	USIM card 1 data signal line	External pull-up 4.7K resistor required
12	USIM1_RST	DO	Reset signal line for USIM card 1	
13	USIM1_CLK	DO	USIM card 1 clock signal line	
14	USIM1_VDD	PO	Power supply for USIM card 1	Automatically identify 1.8V or 3V
79	USIM1_DET	DI	USIM card hot-swappable detection	If not in use, it will be suspended
65	USIM2_VDD	PO	Power supply for USIM card 2	Automatically identify the 1.8V SIM card
62	USIM2_CLK	DO	USIM card 2 clock	

			signal line	
64	USIM2_DATA	IO	USIM card 2 data signal line	External pull-up 4.7K resistor required
63	USIM2_RST	DO	USIM card 2 reset signal line	
Module on/off and Reset				
Pin Number	**Pin Definition**	**IO**	**Function Description**	**Remarks**
7	PWRKEY	DI	Module on/off signal	
15	RESET_N	DI	Module reset signal	Low level effective
USB_BOOT Interface				
Pin Number	**Pin Definition**	**IO**	**Function Description**	**Remarks**
82	BOOT	DI	Forced download and startup	Low level is effective. Do not pull down before the module is powered on
I2C Interface				
Pin Number	**Pin Definition**	**IO**	**Function Description**	**Remarks**
66	I2C_SDA	IO	I2C bus data	1.8V level An external pull-up resistor is required
67	I2C_SCL	DO	I2C bus clock	1.8V level An external pull-up resistor is required
ADC Interface				
Pin Number	**Pin Definition**	**IO**	**Function Description**	**Remarks**
9	ADC0	AI	12-bit universal analog-to-digital conversion	The input range is 0 to 1.2V
96	ADC1	AI	12-bit universal analog-to-digital conversion	The input range is 0 to 1.2V

Radio Frequency Interface				
Pin Number	**Pin Definition**	**IO**	**Function Description**	**Remarks**
35	ANT_MAIN	AIO	LTE&WIFISCAN Antenna	50 Ω characteristic impedance
Main Serial Port				
Pin Number	**Pin Definition**	**IO**	**Function Description**	**Remarks**
17	MAIN_RXD	DI	Main serial port data reception	1.8V level. If not in use, it remains suspended
18	MAIN_TXD	DO	Main serial port data transmission	
22	MAIN_CTS	DO	DTE clearance sending	Connect to the CTS.1.8V level of the DTE. If not in use, leave it floating
23	MAIN_RTS	DI	DTE request is sent	The RTS.1.8V level connected to the DTE is left floating if not in use
Interrupt Wake-up Interface				
19	MAIN_DTR	DI	Used for the host wake-up module	1.8V level. If not in use, it remains suspended
20	MAIN_RI	DO	Used for the module to wake up the host	1.8V level. If not in use, it remains suspended
Auxiliary Serial Port				
Pin Number	**Pin Definition**	**IO**	**Function Description**	**Remarks**
28	AUX_RXD	DI	Auxiliary serial port data reception	1.8V. If not in use, it will be suspended
29	AUX_TXD	DO	Assist in serial port data transmission	1.8V. If not in use, it will be suspended
USB Interface				

Pin Number	Pin Definition	IO	Function Description	Remarks
59	USB_DP	AIO	USB bus differential positive signal	90 Ω differential impedance
60	USB_DM	AIO	USB bus differential negative signal	90 Ω differential impedance
61	USB_VBUS	AI	USB insertion detection	3V~5.25V
Other Interfaces				
Pin Number	**Pin Definition**	**IO**	**Function Description**	**Remarks**
3-6，8，26，44，49-58，68-69，74-78，80-81，83-87，97-109			RESERVED	Stay suspended

NOTE

The IO pin level of this module is 1.8V (except for USIM1, the USIM1 pin level supports 1.8V and 3.0V).

3.3 Power Interface

YM310 A19 module power interface consists of three parts:

- ✧ VBAT is the main power supply for module operation
- ✧ USIM_VDD supplies power to the USIM card
- ✧ VDD_EXT is a 1.8V voltage output power supply

3.3.1 Power Supply Design

The power interface definition of the YM310 A19 module is as follows:

Table 3-4 Definitions of Power Pins

Pin number	Name	I/O	Description	Minimum value	Typical voltage	Maximum value
14	USIM1_VDD	PO	SIM card 1 power supply	0V	1.8V/2.85V	1.98/3.3V

65	USIM2_VDD	PO	SIM card 2 power supply	0V	1.8V	1.98
24	VDD_EXT	PO	1.8V voltage output		1.8V	
42，43	VBAT	PI	Module main power supply	3.3V	3.7V	4.2V

When the module is transmitting at its maximum power, it may cause a significant instantaneous voltage drop on the power supply, resulting in low voltage or insufficient supply current. As a result, the module may shut down or restart. Therefore, to reduce the power fluctuation during the operation of the module, a voltage stabilizing capacitor with a low ESR value should be used. The VBAT traces should be as short and wide as possible to reduce the equivalent impedance of the traces. In addition, to ensure the stability of the power supply, it is recommended to add a TVS diode with VRWM=4.7V, low clamping voltage and peak pulse current IPP at the front end of the power supply.

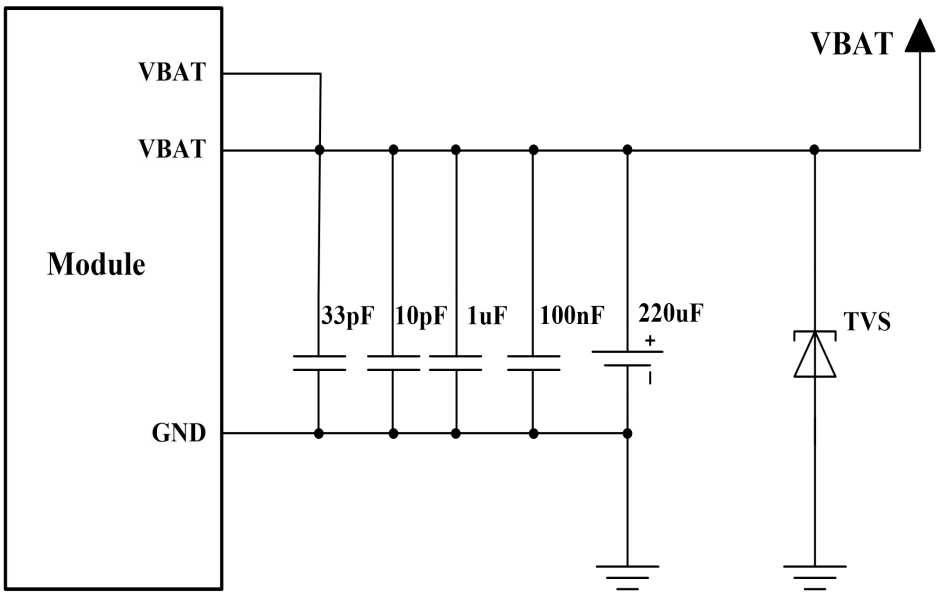


Figure 3-2 Design of Power Supply

Table 3-5 Power Supply Design Description

Recommendation value	Application Instructions	Remarks
220uF	Voltage stabilizing capacitor	Low ESR value capacitors are adopted to reduce power supply fluctuations

WS4.5D3HV	Low-capacitance TVS diode	Avoid power surges or ESD damage to the chip
1uF，100nF	Filter capacitor	Filter out the interference of digital signal noise
33pF，10pF	Filter capacitor	Filter out the radio frequency interference in the low and medium frequency bands

3.3.2 Power Supply Reference Circuit

In actual design, the power supply can be designed using a switching DC power supply or a linear LDO power supply, and then a PMOS transistor is used to control the power supply input so that the power can be completely cut off. Both design circuits need to have sufficient current-carrying capacity. Please refer to the following circuit design for details:

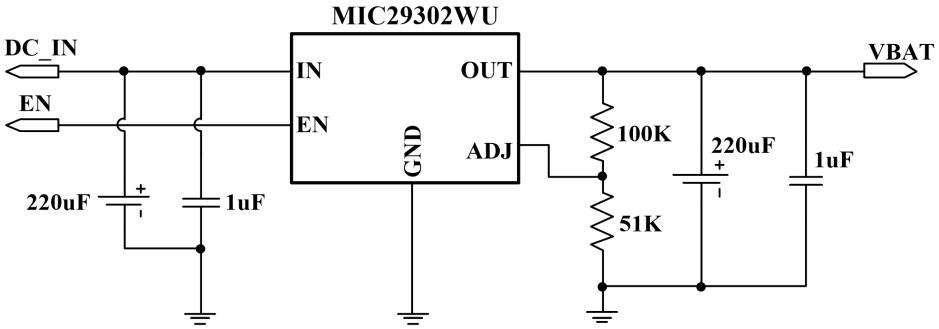


Figure 3-3 LDO linear power supply reference circuit

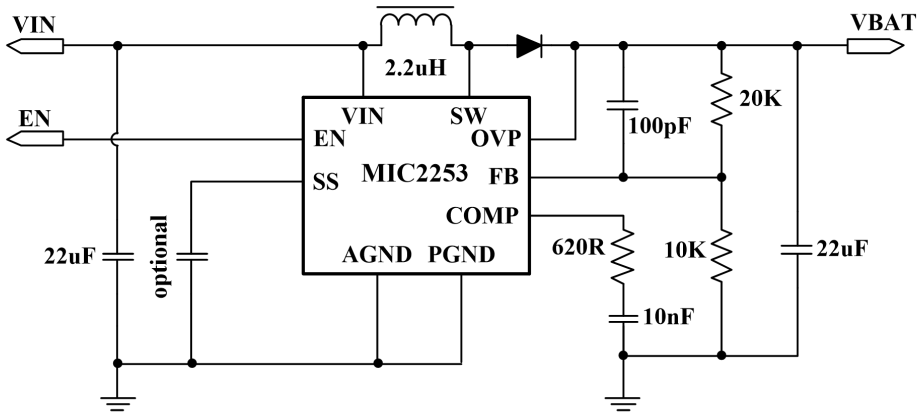


Figure 3-4 Reference circuit of DC Switching Power supply

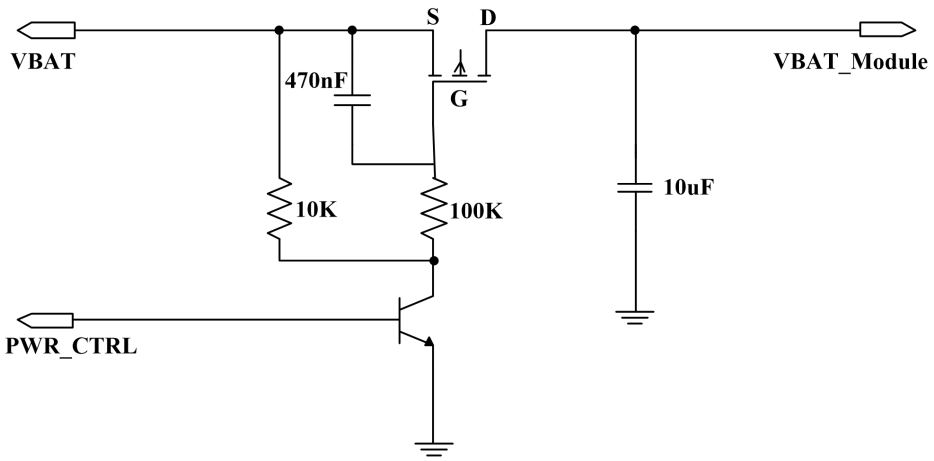


Figure 3-5 Reference circuit for power switch controlled by PMOS transistor

NOTE

- ✧ The minimum operating voltage of the module is 3.3V. Due to the peak current of up to 1.5A generated during data transmission or calls, there will be approximately 600mA of continuous current at the maximum transmission power of the FDD. Therefore, the power supply must be able to provide sufficient current-carrying capacity; otherwise, a significant ripple voltage drop will occur on the power supply voltage, causing the module to restart or operate abnormally.
- ✧ Due to the large current consumption of the module power supply, it is recommended that the power supply PCB traces be as short and wide as possible to minimize the equivalent impedance of the VBAT traces.
- ✧ When the module is in an abnormal state, it is recommended to turn off the module by disconnecting the power supply and then restart it by powering on.

3.3.3 VDD_EXT Voltage Output

After the YM310 A19 module is powered on, it will output a 1.8V voltage through PIN24. This voltage is the logic level voltage of the module. The external master controller can read the voltage of VDD_EXT to determine whether the module is turned on. This voltage can also be used in external circuits with small currents (≤50mA). For example: level conversion chips, GPIO pull-up, etc. If not in use, keep it suspended.

3.4 Reset Control

The PIN15 of the YM310 A19 module is the reset pin. When the application end detects an abnormality in the module or the software is unresponsive, the module can be reset. Pull this pin down for at least 300ms to complete the reset. The RESET signal is relatively sensitive to interference. A capacitor no larger than 10nF can be reserved near the signal for signal filtering, and the wiring should be kept away from the RF interference signal.

Table 3-6 Definition of Reset Pins

Pin	Signal Name	I/O Attribute	High Level Value	Description
15	RESET_N	DI	1.8V±0.3V	Low Level Effective

Table 3-7 Reset Methods

Reset Mode	Reset Mode
AT Command Reset	AT+CFUN=1,1
Hardware Reset	Pulling down the RESET_N pin for at least 300ms and then releasing it can reset the module

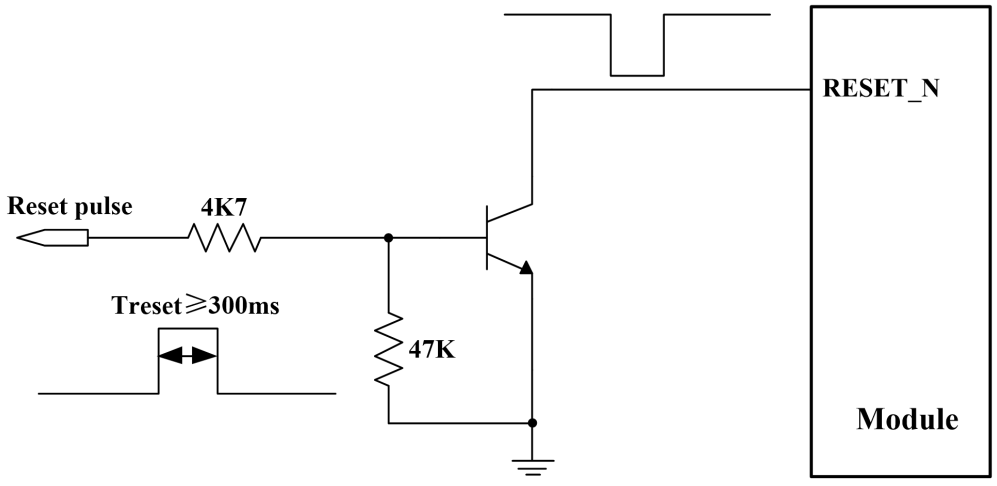


Figure 3-6 Reset reference circuit

Table 3-8 RESET Pin Parameters

Symbol	Description	Minimum value	Typical value	Maximum value	Unit
Treset	Low-level pulse	300		-	ms

	width				
VIH	RESET inputs a high-level voltage		1.8	2.1	V
VIL	RESET input low-level voltage	-0.3	0	0.8	V

The RESET sequence is as follows:

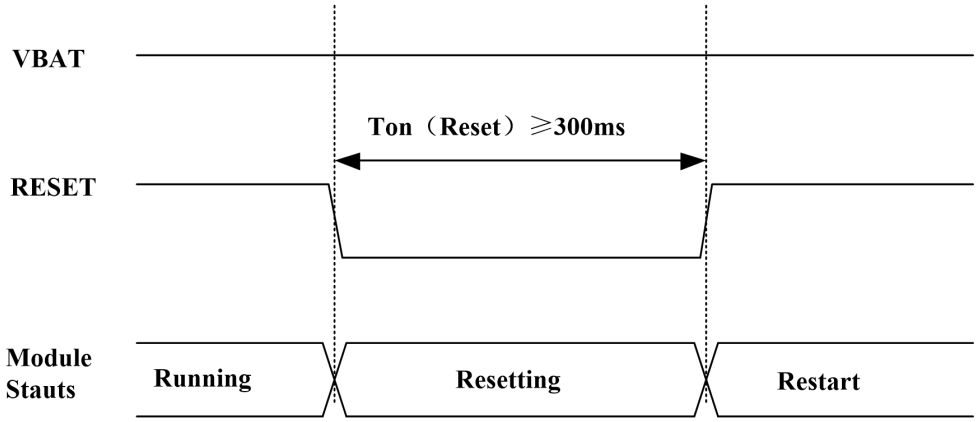


Figure 3-7 Reset timing diagram

The YM310 A19 module supports AT command reset. The AT command is AT+CFUN=1, and 1 can restart the module. Detailed instructions can be found in the AT instruction set manual.

3.5 Module Startup

The PIN7 pin of the YM310 A19 module is the power-on pin. The module can be powered on by pulling down the PWRKEY of the PIN7 pin for at least 500ms. Users can determine whether the module is powered on by querying the high and low levels of the VDD_EXT pin.

Table 3-9 Definitions of Power on/Off Pins

Pin	Signal Name	I/O Attribute	High-Level Value	Description
7	PWRKEY	DI	VBAT	Low level Effective

The startup sequence is as follows:

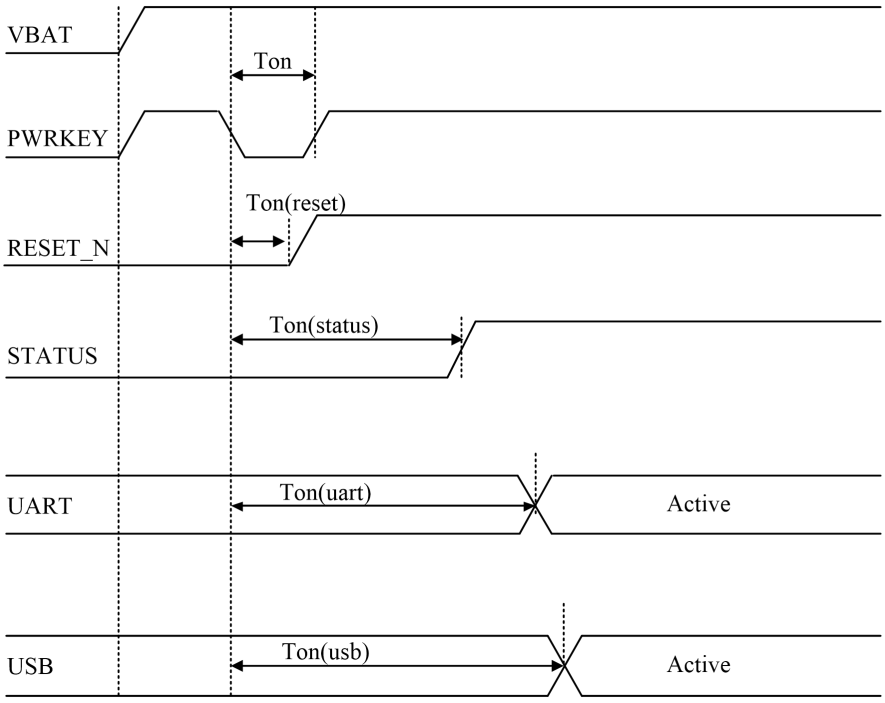


Figure 3-8 Startup sequence diagram

Table 3-10 Startup Timing Parameters

Symbol	Description	Minimum value	Typical Value	Maximum value	Unit
Ton	Startup Low Level Width	500	700	-	ms
Ton(reset)	Startup time (determined by the reset status)		TBD	-	s
Ton(status)	Startup time (determined by status)		TBD	-	s
Ton(uart)	Startup time (judged by the status of uart)		TBD	-	s
Ton(usb)	Startup time (determined by usb status)		TBD	-	s
VIH	PWRKEY input high level	0.6	3.8	4.2	v
VIL	The input level of PWRKEY is low	-0.3	0	0.5	v

It is recommended to use an open-set driver circuit to control the PWRKEY. Release it after

raising the base level for at least 500ms, at which point the module is powered on. The power can also be turned on and off by pressing a button. A TVS diode needs to be placed near the button for ESD protection.

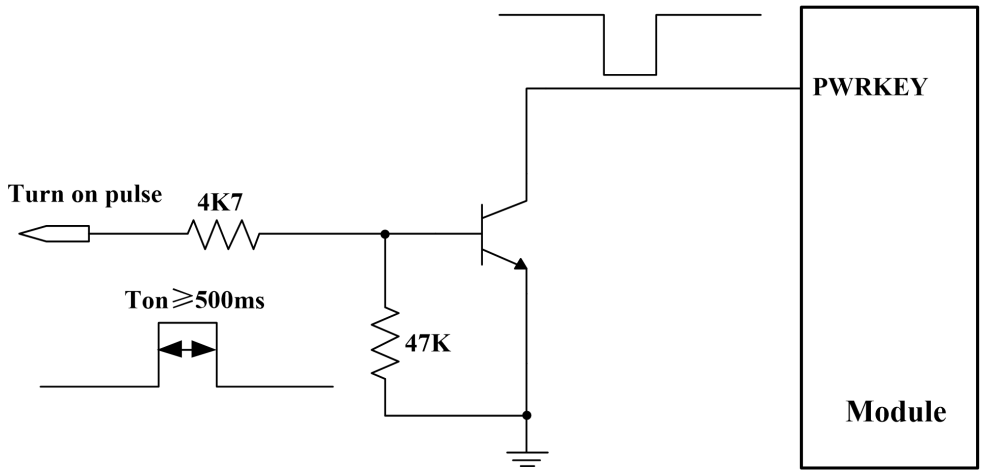


Figure 3-9 Open set driver startup reference circuit

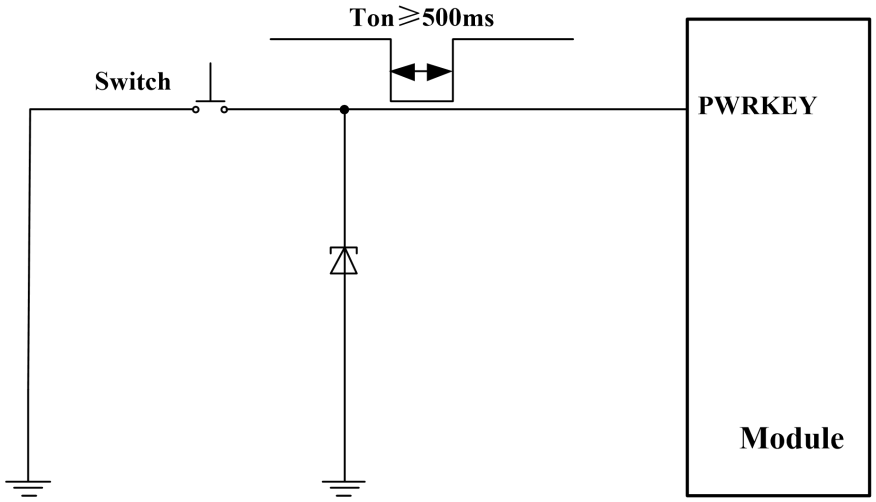


Figure 3-10 Reference circuit for key startup

NOTE

- ✧ The module boots up in a low-level mode by default.
- ✧ The power-on automatic power-on mode can be achieved by directly pulling down the PWRKEY pin of the module to the ground. The pull-down resistor is recommended to be 4.7KΩ. This startup method does not support module key shutdown.

✧ If you need a low-pulse startup mode, you can contact your local FAE for support.

3.6 Module Shutdown

The YM310 A19 module supports the following three shutdown methods.

Table 3-11 Module Shutdown Methods

Shutdown Mode	Shutdown Conditions	Description
Low-voltage Shutdown	The power supply voltage is too low or there is an abnormal power failure	The module did not go through the normal shutdown process
Hardware Shutdown	Pull down the PWRKEY pin for more than 3 seconds	Perform the normal shutdown process
AT Command Shutdown	AT Command	Software Shutdown

When the module is working normally, it cannot be shut down by cutting off the power supply, as this may damage the Flash data of the module. It is recommended to execute the shutdown process through the power on/off pin or the AT command.

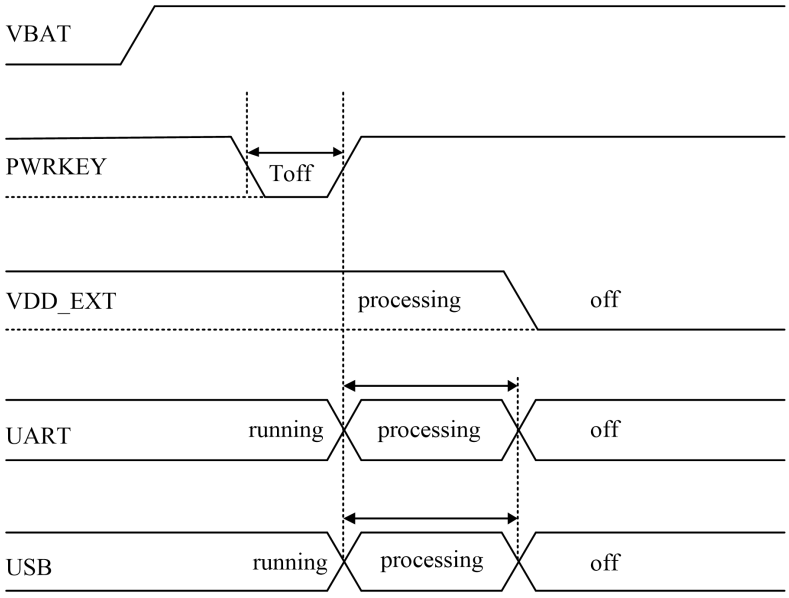


Figure 3-11 Shutdown sequence diagram

3.7 Forced Download

YM310 A19 supports the BOOT function. Before the module is turned on, the BOOT pin can be pulled to GND. When the module is turned on, it will enter a forced download mode. At this time, the software of the module can be upgraded through the USB interface.

Table 3-12 Pin Definitions of the USB_BOOT interface

Pin Number	Pin Definition	IO	Function Description
82	BOOT	DI	Forced download and startup

3.8 USB Interface

The YM310 A19 module supports one USB2.0 interface, supports device mode, but does not support USB charging function. The USB cable must comply with the USB2.0 protocol specification. The USB interface is defined as follows:

Table 3-13 USB Interface Pin Definitions

Pin Number	Signal Name	IO	Description
59	USB_DP	IO	USB bus differential positive signal
60	USB_DM	IO	USB bus differential negative signal
61	USB_VBUS	AI	USB insertion detection

The module, as a USB slave device, supports USB sleep and wake-up mechanisms. The reference circuit for the USB interface application is as follows:

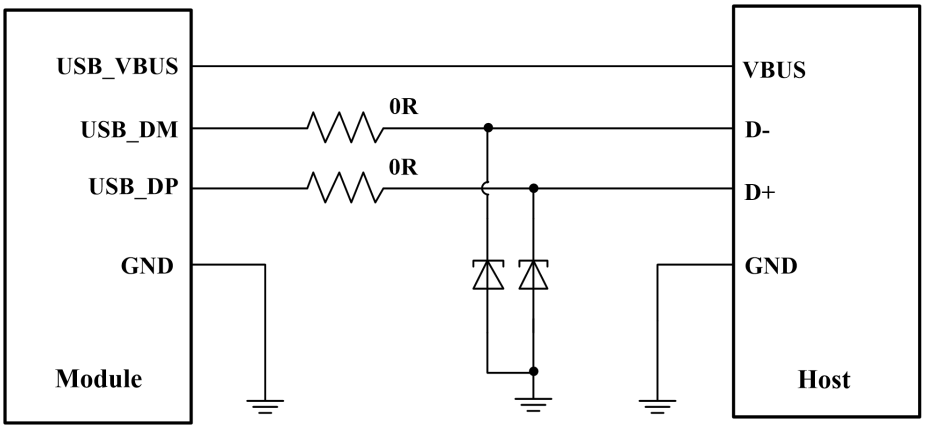


Figure 3-12 USB connection design circuit diagram

NOTE

- ✧ USB supports high-speed (480Mbps) and full-speed (12Mbps) modes. The cable design must strictly follow the requirements of the USB2.0 protocol, paying attention to the protection of data cables. Differential cables should be used, the shorter the better, and kept as far away as possible from high-speed signals or other signals of the same frequency. The control impedance should be 90Ω.
- ✧ To enhance the anti-static performance of the USB interface, it is recommended to add ESD protection devices to the data cable, with the equivalent capacitance value of the protection device being less than 1pF. It is recommended to connect a 0-ohm resistor in series on the data cable.
- ✧ The USB interface of the module does not provide USB bus power externally. The module can only be used as a slave device.
- ✧ The functions supported by the USB interface include: software download and upgrade, data communication, AT Command, etc.

3.9 UART Interface

The YM310 A19 module provides three sets of UART interfaces. Among them, one group is the main serial port, one group is the debugging serial port, and one group is the two-wire auxiliary serial port.

3.9.1 Serial Port

Main Serial Port:

This serial port can realize AT interaction instructions, data interaction with peripheral devices, etc.

The baud rate of the main serial port of the module can be set to 9600,19200,38400,57600,115200,230400,460800,921600 bps, etc. The default is 115200bps.

The main serial port interface is defined as follows:

Table 3-14 Definition of Main Serial Port Signal

Pin Number	Signal Name	Attribute	Description
17	MAIN_RXD	DI	Main serial port data reception

18	MAIN_TXD	DO	Main serial port data transmission
22	MAIN_CTS	DO	DTE clearance sending
23	MAIN_RTS	DI	DTE request is sent

The auxiliary serial port interface is defined as follows:

Table 3-15 Definition of Auxiliary Serial Port Interface

Pin Number	Signal Name	Attribute	Description
28	AUX_RXD	DI	Auxiliary serial port data reception
29	AUX_TXD	DO	Assist in serial port data transmission

When using the serial port function, you can refer to the following connection methods:

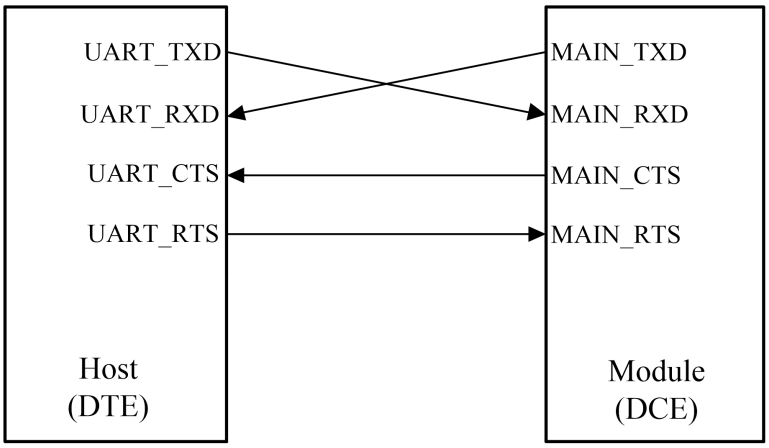


Figure 3-13 Serial port design drawing

If the module serial port is connected to an MCU with a 3.3V level, since the level of the module serial port is 1.8V, a level conversion chip is needed to achieve level matching. The connection method of the chip can be referred to the following circuit:

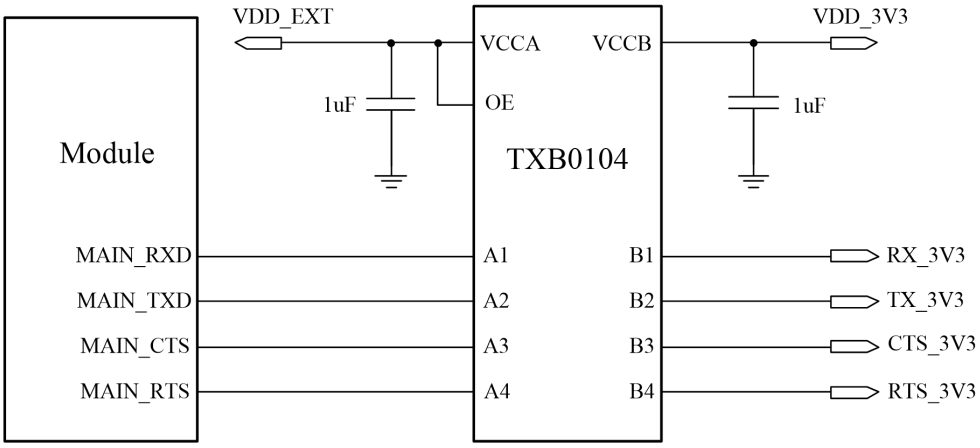


Figure 3-14 Level conversion chip circuit

NOTE

✧ It is necessary to pay attention to the input and output directions of the CTS and RTS pins of the serial port hardware flow control.

3.9.2 Debug the Serial Port

The YM310 A19 module provides a 2-wire DEBUG serial port for use as a debugging module. The debugging serial port supports a baud rate of 115200bps, which is used for Linux control and log printing. Test points can be reserved. If not in use, please keep them floating.

Table 3-16 Definitions of Debug Serial Port Pins

Pin Number	Signal Name	Attribute	Description
38	DBG_RXD	DI	Module receives data.
39	DBG_TXD	DO	Module sends data

3.10 Sleep Wake-up Interface

The YM310 A19 module supports the sleep wake-up function. Through sleep, the power consumption of the module can be reduced. The way to wake up from hibernation can be achieved through different interface communication methods. For details, please refer to the following content.

3.10.1 UART Interface Communication Method

When the host and the module communicate through the UART interface, please refer to the following figure for the connection of their sleep and wake-up hardware:

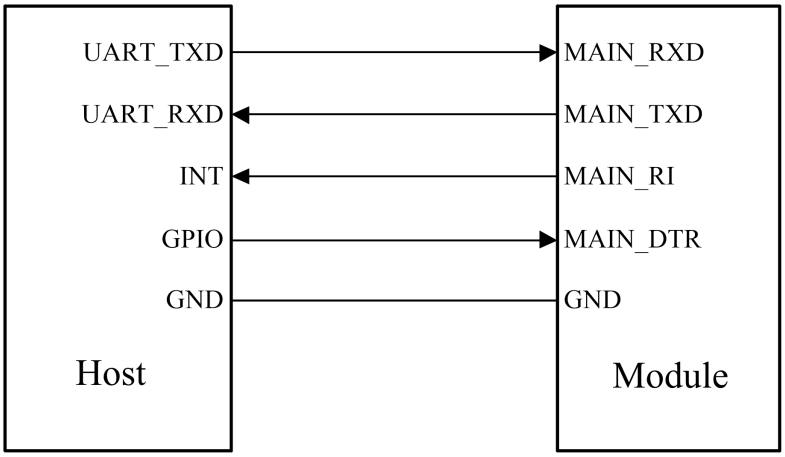


Figure 3-15A UART interface sleep wake-up hardware connection diagram

Simultaneously satisfy the following two conditions:

1.MAIN_DTR has been at a high level or left floating all the time;

2. Execute the AT command: AT+CSCLK=1. The module then enters the sleep mode.

The module is awakened by pulling down the MAIN_DTR of the module through the host. When the module has URC to report, the module can wake up the host through MAIN_RI.

3.10.2 USB Interface Communication Method

When the host and the module communicate through the USB interface, their sleep and wake-up can be divided into the following three situations:

3.10.2.1 Supports USB suspension and wake-up as well as USB remote wake-up functions

For the hardware connection of the host supporting USB suspension and wake-up as well as USB remote wake-up functions, please refer to the following figure:

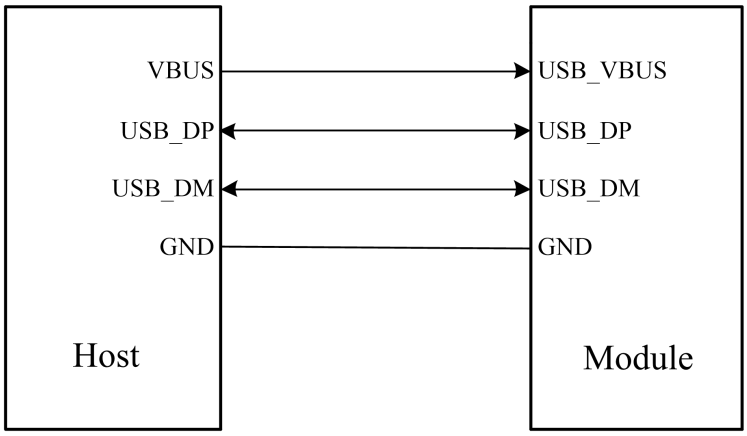


Figure 3-15B shows the hardware connection diagram supporting USB remote wake-up function

Simultaneously satisfy the following three conditions:

1. MAIN_DTR has been at a high level or left floating all the time;

2. Execute the AT command: AT+CSCLK=1;

3. The USB bus of the host connected to the USB interface of the module enters the suspended state. The module then enters the sleep mode.

The host can send data to the module via USB to wake it up. When the module has URC to report, the module can send remote wake-up data via USB bus to wake up the host.

3.10.2.2 Supports USB suspension and wake-up as well as RI signal wake-up functions

If the host supports USB suspension and wake-up but does not support the USB remote wake-up function, the host can be awakened through the module MAIN_RI signal. For specific hardware connections, please refer to the following figure:

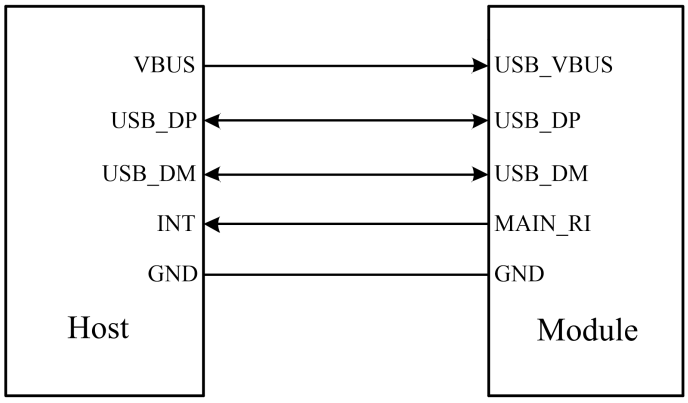


Figure 3-15C Hardware connection diagram supporting MAIN_RI wake-up function

Simultaneously satisfy the following three conditions:

1. MAIN_DTR has been at a high level or left floating all the time;

2. Execute the AT command: AT+CSCLK=1;

3. The USB bus of the host connected to the USB interface of the module enters the suspended state. The module then enters the sleep mode.

The host can send data to the module via USB to wake up the module. After setting the MAIN_RI working mode through the AT command, when the module has URC to report, the module can first send data through the MAIN_RI signal pin to wake up the host. After the host is awakened, it reads the URC.

3.10.2.3 It does not support USB suspension, module sleep and wake-up functions

If the host does not support USB suspension, please refer to the following figure for the hardware connection between the host and the module:

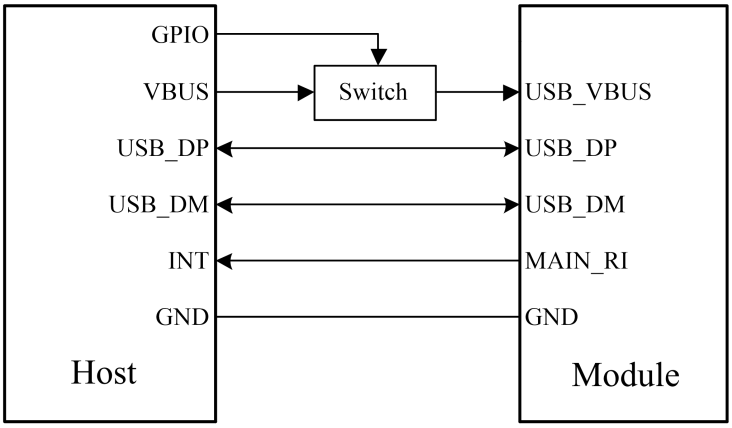


Figure 3-15D shows the hardware connection diagram that does not support the USB suspension function

Simultaneously satisfy the following three conditions:

1. MAIN_DTR has been at a high level or has remained floating all the time;

2. Execute the AT command: AT+CSCLK=1;

3. Disconnect the power supply of USB_VBUS. The module then enters the sleep mode;

The host can control the power supply switch of USB_VBUS through the GPIO port and wake up the module through the power supply of USB_VBUS.

3.11 USIM Interface

The YM310 A19 module provides two USIM card interfaces compatible with the ISO 7816-3 standard. The power supply of the USIM card is provided by the internal power manager of the module. USIM1 supports voltages of 1.8V/3.0V, while USIM2 only supports 1.8V.

Table 3-17 Definition of SIM Card Signal

Pin Number	Signal Name	Attribute	Description
11	USIM1_DATA	IO	USIM card 1 data signal line
12	USIM1_RST	DO	Reset signal line for USIM card 1
13	USIM1_CLK	DO	USIM card 1 clock signal line
14	USIM1_VDD	PO	Power supply for USIM card 1
79	USIM1_DET	DI	USIM card hot-swappable detection
65	USIM2_VDD	PO	Power supply for USIM card 2
62	USIM2_CLK	DO	USIM card 2 clock signal line
64	USIM2_DATA	IO	USIM card 2 data signal line
63	USIM2_RST	DO	USIM card 2 reset signal line

The YM310 A19 module does not come with a USIM card slot. When in use, users need to design a USIM card slot on their own interface board. The reference circuit of the USIM card interface is as follows:

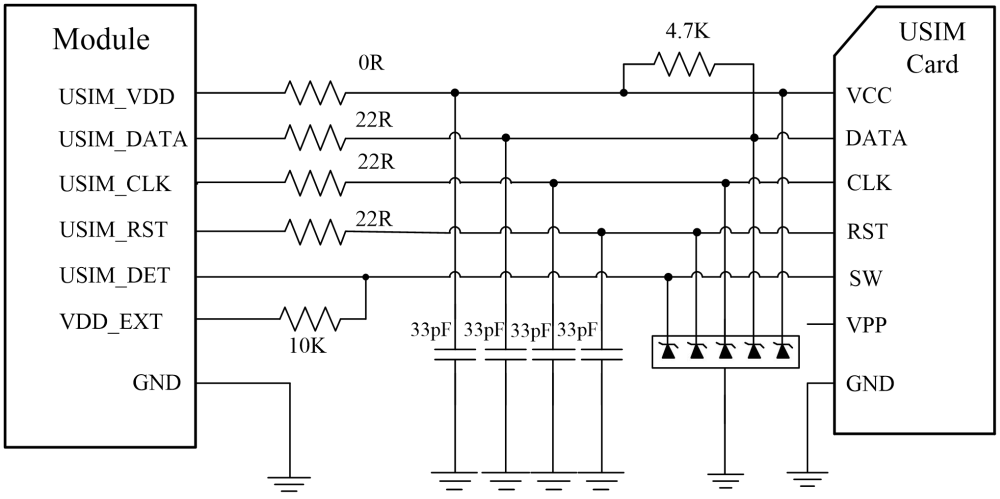


Figure 3-16 USIM design circuit diagram

NOTE

- ✧ The peripheral circuit devices of the USIM interface should be placed close to the card slot, and the USIM card slot should be arranged close to the module.
- ✧ The circuit of a USIM card is susceptible to radio frequency interference, which can cause the card to become unrecognizable or dropped. Therefore, the card slot should be placed as far away from the antenna radio frequency radiation as possible, and the card wiring should be kept away from radio frequency, power supply, and high-speed signal lines as much as possible.
- ✧ To prevent instantaneous voltage overload for the USIM interface, it is recommended to connect a 22R resistor in series on each signal line path.
- ✧ The ground of the USIM card slot and the ground of the module should maintain good connectivity.
- ✧ The USIM.DET pin can set the detection function through AT command according to different card sockets. For example, when using a normally closed USIM card socket, set AT+HOSCFG=1,1 and set the status to high when the USIM card is in place; When using a normally open USIM card slot, set AT+HOSCFG=1,0 to low when the USIM card is in place, and set AT+HOSCFG=0,0 to disable the SIM card hot plug function.
- ✧ The hot plugging function is only supported by USIM1, not by USIM2.

3.12 Status Instruction Interface

YM310 A19 module provides two GPIO pins to indicate the module status.

Table 3-18 Definition of Status Indicator Pins

Pin	Signal Name	I/O Attribute	Description
16	NET_STATUS	DO	Module network status indication
25	STATUS	DO	Module operation status indication

Table 3-19 Module Operating Status Indication

Module Operation Status Indication	Pin Level
Power on Status	high Level
Other	Low Level

Table 3-20 Module Network Status Indication

Network Operation Status Indication	Pin Level
During the Phone Call	High level
Data Transmission Status	Flash (125ms high /125ms low)
Standby State	Slow flash (1800ms high /200ms low)
Search Status	Slow flash (200ms high /1800ms low)

The reference design drawing of the module network status indicator light is as follows:

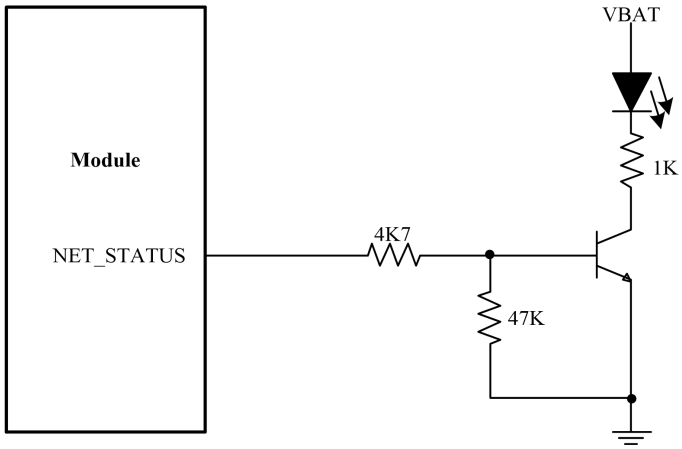


Figure 3-17 Circuit diagram of network status indicator light

NOTE

✧ The brightness of the network indicator light can be adjusted by regulating the current-limiting resistor.

3.13 I2C Bus

The YM310 A19 module provides a set of I2C interfaces. The I2C interface has a 1.8V level value, a 5.0 protocol interface, and a clock rate of 400KHz.

Table 3-21 Definition of I2C pins

Pin	Signal Name	I/O Attribute	Description
66	I2C2_SDA	IO	I2C bus data
67	I2C2_SCL	DO	I2C bus clock

The connection method of the I2C reference circuit is as follows:

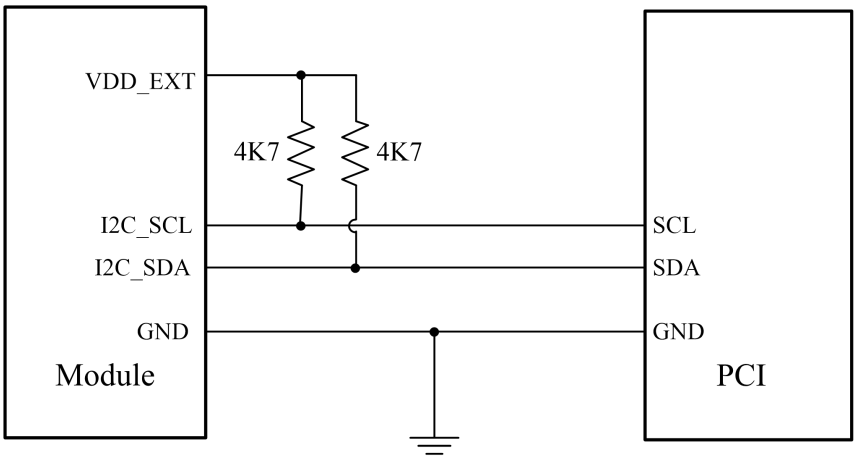


Figure 3-18 Reference circuit diagram of I2C interface

3.14 PCM Voice Interface

The YM310 A19 module provides a set of PCM digital audio interfaces to enable communication with external CODE or DAC audio devices.

Table 3-22 PCM/PWM Pin Definitions

Pin	Signal Name	I/O Attribute	Description
30	PCM_CLK	DO	PCM Clock Pulse
31	PCM_SYNC	DO	PCM frame synchronization signal
32	PCM_OUT	DO	PCM sends data
	PWM	AO	Pulse width modulation signal
33	PCM_IN	DI	PCM receives data

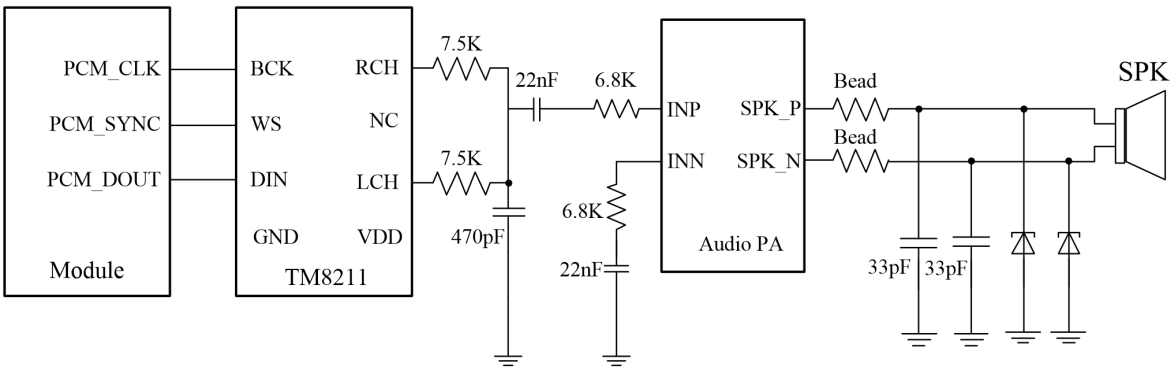


Figure 3-19 DAC audio reference circuit

In addition, the PCM_OUT signal multiplexes the PWM function output to achieve the audio playback function, without the need for an external Codec chip. It is suitable for application

scenarios where the requirements for sound quality are not high but cost is sensitive.

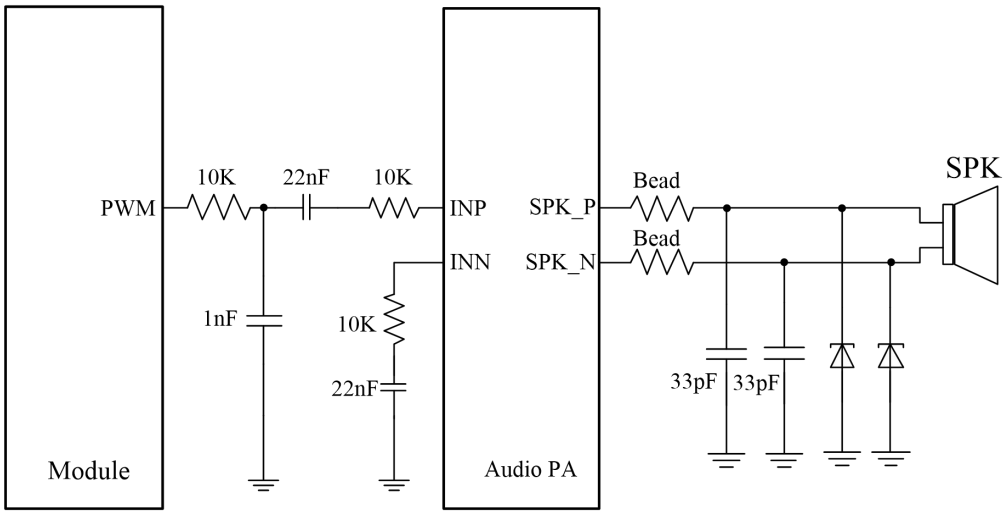


Figure 3-20 PWM audio reference diagram

3.15 ADC Interface

The YM310 A19 module provides two analog-to-digital converter interfaces to read voltage values. The input voltage of the ADC interface should not exceed 1.2V. It is recommended to use a voltage divider circuit for the ADC pin input.

Table 3-23 ADC Pin Definitions

Pin Number	Signal Name	Description	Level Value (V)			Remarks
			Minimum value	Typical value	Maximum value	
9	ADC0	Analog-to-digital converter interface	0		1.2V	12-bit accuracy
96	ADC1					

3.16 Radio Frequency Interface

The YM310 A19 module provides one LTE antenna interface, which is responsible for the reception and transmission of the module's RF signals. The characteristic impedances of the antenna interfaces are all 50Ω.

Table 3-24 Antenna Interface Pin Definitions

Pin number	Signal Name	I/O attribute	Description	Remarks
35	ANT_MAIN	AIO	LTE antenna	50 Ω

			interface	characteristic impedance

3.16.1 Antenna Matching Circuit

To facilitate antenna debugging, it is recommended that a π -shaped matching circuit be added between the module signal port and the feed point of the mainboard for the antenna interface, and a 50-ohm impedance line be used.

The circuit is as shown in the following figure:

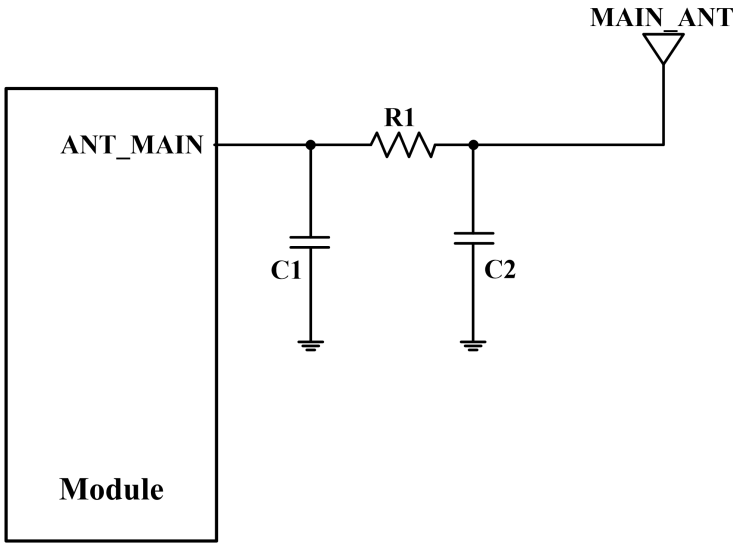


Figure 3-21 Antenna Matching Circuit

NOTE

- ✧ The antenna interfaces of the YM310 A19 module are all led out by solder pads. Therefore, when designing, matching RF connection lines need to be adopted.
- ✧ In actual design, users can adjust and match the device parameter values according to the wiring of the circuit board by the antenna factory. The motherboard R1 defaults to 0 ohms, and C1/C2 defaults to empty patches.
- ✧ The antenna is a sensitive device that is easily affected by the external surrounding environment. Therefore, it needs to be kept away from interference signals such as digital clock lines and DC power supplies. It is recommended to use a complete ground layer as the reference site.
- ✧ The antenna LAYOUT should have as short and straight traces as possible, avoiding through holes and layering. It should be three-dimensional and grounded, and more holes should be added on both sides of the trace for isolation.

3.16.2 RF Wiring Reference

The antenna of the YM310 A19 module is led out using solder pads, and microstrip lines or other types of RF wiring must be used from the antenna solder pad to the antenna feed point. The characteristic impedance of the signal line should be controlled at 50 Ω.

The impedance of RF signal lines is determined by the dielectric constant of the material, the wire width (W), the ground gap (S), and the height of the reference ground plane (H). Therefore, impedance simulation tools are required to calculate the impedance value of RF wiring.

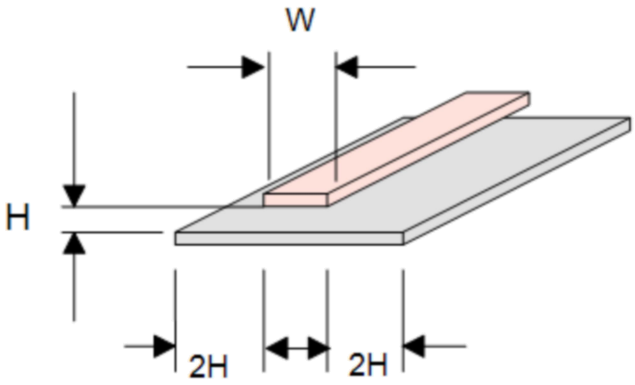


Figure 3-22 Complete structure of microstrip line

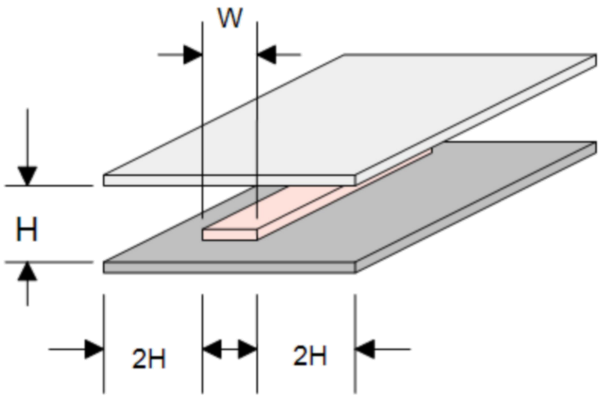


Figure 3-23 Complete structure of strip line

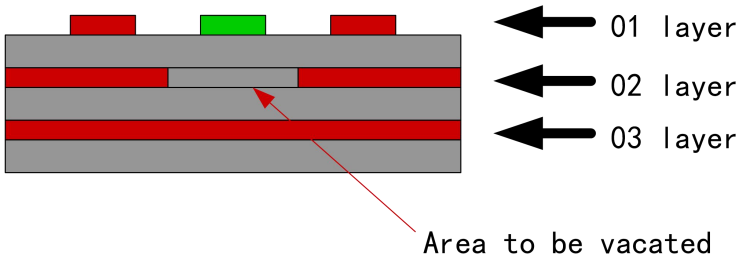


Figure 3-24 shows the structure of the third-layer PCB microstrip transmission line as a reference

Chapter 4 Overall technical indicators

4.1 Overview of This Chapter

The overall technical specifications of the YM310 A19 module RF include the following parts:

- ✧ Working Frequency
- ✧ Radio Frequency Conduction Measurement
- ✧ Conduction receiving sensitivity and transmission power
- ✧ Antenna Requirements
- ✧ Module Power Consumption Characteristics

4.2 Working Frequency

Table 4-1 4G Frequency Table

Frequency band	Up Frequency	Downlink Frequency	Duplex Mode
LTE B1	1920MHz–1980MHz	2110MHz–2170MHz	FDD
LTE B2	1850MHz–1910MHz	1930MHz–1990MHz	FDD
LTE B3	1710MHz–1785MHz	1805MHz–1880MHz	FDD
LTE B4	1710MHz–1755MHz	2110MHz–2155MHz	FDD
LTE B5	824MHz–849MHz	869MHz–894MHz	FDD
LTE B7	2500MHz – 2570MHz	2620MHz – 2690MHz	FDD
LTE B8	880MHz–915MHz	925MHz–960MHz	FDD
LTE B12	699MHz – 716MHz	729MHz – 746MHz	FDD
LTE B13	777MHz – 787MHz	746MHz – 756MHz	FDD
LTE B17	704MHz – 716MHz	734MHz – 746MHz	FDD
LTE B18	815MHz – 830MHz	860MHz – 875MHz	FDD
LTE B19	830MHz – 845MHz	875MHz – 890MHz	FDD
LTE B20	832MHz – 862MHz	791MHz – 821MHz	FDD
LTE B26	814MHz – 849MHz	859MHz – 894MHz	FDD
LTE B28	703MHz – 748MHz	758MHZ – 803MHz	FDD
LTE B66	1710MHz – 1780MHz	2110MHz – 2180MHz	FDD

LTE B71	663MHz – 698MHz	617MHz – 652MHz	FDD
LTE B38	2570MHz–2620MHz	2570MHz–2620MHz	TDD
LTE B40	2300MHz–2400MHz	2300MHz–2400MHz	TDD
LTE B41	2496MHz–2690MHz	2496MHz–2690MHz	TDD

4.3 Radio Frequency Conduction Measurement

4.3.1 Test Environment

Table 4-2 Testing Instruments

Testing Instrument	Power supply	Murata Coaxial RF Cable
R&S CMW500	Agilent 66319	MXHP32HP1000

4.3.2 Test Standard

The YM310 A19 module passed the test standards of 3GPP TS 51.010-1, 3GPP TS 34.121-1, and 3GPP TS 36.521-1. Each module is strictly tested in the factory to ensure reliable quality.

4.4 Conduction Receiving Sensitivity and Transmission Power

The test indicators for 4G receiving sensitivity and transmitting power of YM310 A19 module are as follows:

Table 4-3 4G Radio Frequency Sensitivity Index

Directory (Sensitivity)	Requirements of the 3GPP agreement	small est	Typical	Largest
LTE B1(FDD QPSK Pass >95%)	< - 94(10MHz)		-100.2	-99.7
LTE B2(FDD QPSK Pass >95%)	< - 92(10MHz)		-100.2	-99.7
LTE B3(FDD QPSK Pass >95%)	< - 91(10MHz)		-99.7	-99.2
LTE B4(FDD QPSK Pass >95%)	< - 94(10MHz)		-99.7	-99.2
LTE B5(FDD QPSK Pass >95%)	< - 92.5(10MHz)		-99.3	-99.2
LTE B7(FDD QPSK Pass >95%)	< - 92(10MHz)		-97.7	-97.2
LTE B8(FDD QPSK Pass >95%)	< - 91.5(10MHz)		-99.7	-99.2

LTE B12(FDD QPSK Pass>95%)	< - 91(10MHz)		-97.7	-97.2
LTE B13(FDD QPSK Pass >95%)	< - 91(10MHz)		-98.2	-97.7
LTE B17(FDD QPSK Pass >95%)	< - 91(10MHz)		-97.7	-97.2
LTE B18(FDD QPSK Pass >95%)	< - 94.5(10MHz)		-98.7	-98.2
LTE B19(FDD QPSK Pass >95%)	< - 94.5(10MHz)		-98.7	-98.2
LTE B20(FDD QPSK Pass >95%)	< - 91.5(10MHz)		-99.2	-98.7
LTE B26(FDD QPSK Pass >95%)	< - 92(10MHz)		-99.2	-98.7
LTE B28(FDD QPSK Pass >95%)	< - 93(10MHz)		-98.7	-98.2
LTE B66(FDD QPSK Pass >95%)	< - 93.5(10MHz)		-99.2	-98.7
LTE B71(FDD QPSK Pass >95%)	< - 93.5(10MHz)		-97.2	-97.2
LTE B38(TDD QPSK Pass >95%)	< - 94.5(10MHz)		-98.7	-98.2
LTE B40(TDD QPSK Pass >95%)	< - 94.5(10MHz)		-98.7	-98.2
LTE B41(TDD QPSK Pass >95%)	< - 92.5(10MHz)		-98.7	-98.2

Table 4-4 4G Radio Frequency Transmission Power Index

Directory	Maximum (dBm) of 3GPP protocol	Smallest	Typical	Largest
LTE B1	21 to 25	21	23	24
LTE B2	21 to 25	21	23	24
LTE B3	21 to 25	21	23	24
LTE B4	21 to 25	21	23	24
LTE B5	21 to 25	21	23	24
LTE B7	21 to 25	21	23	24
LTE B8	21 to 25	21	23	24
LTE B12	21 to 25	21	23	24
LTE B13	21 to 25	21	23	24
LTE B17	21 to 25	21	23	24
LTE B18	21 to 25	21	23	24
LTE B19	21 to 25	21	23	24
LTE B20	21 to 25	21	23	24
LTE B26	21 to 25	21	23	24
LTE B28	21 to 25	21	23	24

LTE B66	21 to 25	21	23	24
LTE B71	21 to 25	21	23	24
LTE B38	21 to 25	21	23	24
LTE B40	21 to 25	21	23	24
LTE B41	21 to 25	21	23	24

4.5 Antenna Requirements

YM310 A19 Module antenna design requirements:

Table 4-5 Antenna Index Requirements

Band	VSWR	Insertion loss	Efficiency	TRP	TIS
B1 FDD	<2:1	<1.5db	>40%	>16.5	<-88
B2 FDD	<2:1	<1.5db	>40%	>16.5	<-88
B3 FDD	<2:1	<1.5db	>40%	>16.5	<-88
B4 FDD	<2:1	<1.5db	>40%	>16.5	<-88
B5 FDD	<2:1	<1.0db	>40%	>16.5	<-88
B7 FDD	<2:1	<2.0db	>40%	>16.5	<-88
B8 FDD	<2:1	<1.0db	>40%	>16.5	<-88
B12 FDD	<2:1	<1.0db	>40%	>16.5	<-88
B13 FDD	<2:1	<1.0db	>40%	>16.5	<-88
B17 FDD	<2:1	<1.0db	>40%	>16.5	<-88
B18 FDD	<2:1	<1.0db	>40%	>16.5	<-88
B19 FDD	<2:1	<1.0db	>40%	>16.5	<-88
B20 TDD	<2:1	<1.0db	>40%	>16.5	<-88
B26 TDD	<2:1	<1.0db	>40%	>16.5	<-88
B28 TDD	<2:1	<1.0db	>40%	>16.5	<-88
B66 FDD	<2:1	<1.5db	>40%	>16.5	<-88
B71 FDD	<2:1	<1.0db	>40%	>16.5	<-88
B38 TDD	<2:1	<2.0db	>40%	>16.5	<-88
B40 TDD	<2:1	<2.0db	>40%	>16.5	<-88
B41 TDD	<2:1	<2.0db	>40%	>16.5	<-88

NOTE

The antenna performance is only a recommended value. Customers can evaluate and adjust it with the antenna supplier based on the actual test results and certification requirements to ensure the overall antenna performance of the terminal product.

4.6 Power Consumption Characteristics

Table 4-6 Dormant Idle Power Consumption

Mode	Test Conditions	Current value (Avg)	Unit
Leakage of electricity when shutting down	Module Shutdown	5	uA
Hibernation Mode	LTE-FDD @ DRX =0.32S （USB Disconnected）	1.348	mA
	LTE-FDD @ DRX =0.64S （USB Disconnected）	1.098	mA
	LTE-FDD @ DRX =1.28S （USB Disconnected）	0.939	mA
	LTE-FDD @ DRX =2.56S （USB Disconnected）	0.859	mA
Idle Mode	LTE-B1 @ PF = 64 （ USB Disconnected）	8.6	mA
	LTE-B40 @ PF = 64 （ USB Disconnected）	8.2	mA
	LTE-B1 @ PF = 64 （ USB Connection）	18.7	mA
	LTE-B40 @ PF = 64 （ USB Connection）	18.27	mA

Table 4-7 Power Consumption of LTE Data Transmission

Band	Channel	Power dBm	Current consumption mA
LTE-FDD B1	18050	21.82	495

@10Mhz,FRB	18300	21.86	467
	18550	21.7	491
LTE-FDD B2 @10Mhz,FRB	18650	21.83	462
	18900	21.92	428
	19150	21.83	471
LTE-FDD B3 @10Mhz,FRB	19250	21.83	519
	19575	21.95	480
	19900	21.65	519
LTE-FDD B4 @10Mhz,FRB	20000	22.14	535
	20175	22.26	511
	20350	22.1	483
LTE-FDD B5 @10Mhz,FRB	20450	22.24	447
	20525	22.13	474
	20600	22.13	445
LTE-FDD B7 @10Mhz,FRB	20800	21.7	586
	21100	21.93	552
	21400	21.6	579
LTE-FDD B8 @10Mhz,FRB	21500	22	468
	21625	22	451
	21750	22.1	460
LTE-FDD B12 @10Mhz,FRB	23060	22.25	476
	23095	22.16	475
	23130	22.1	477
LTE-FDD B13 @10Mhz,FRB	-	-	-
	23230	21.8	467
	-	-	-
LTE-FDD B17 @10Mhz,FRB	23780	22.1	471
	23790	22	474
	23800	22.02	476
LTE-FDD B18 @10Mhz,FRB	23900	22.09	502
	23925	21.9	499

	23950	22.08	514
LTE-FDD B19 @10Mhz,FRB	24050	22.19	507
	24075	22.1	498
	24100	22.1	504
LTE-FDD B20 @10Mhz,FRB	24200	22.04	485
	24300	22.2	492
	24400	22.02	484
LTE-FDD B26 @10Mhz,FRB	26740	22.3	517
	26865	22.3	522
	26990	22.1	518
LTE-FDD B28 @10Mhz,FRB	27260	21.97	492
	27435	21.84	462
	27610	21.53	460
LTE-FDD B66 @10Mhz,FRB	132022	22.1	494
	132322	22.16	452
	132622	21.92	488
LTE-TDD B38 @10Mhz,FRB	37800	21.9	246
	38000	21.9	244
	38200	222	243
LTE-TDD B40 @10Mhz,FRB	38700	22	296
	39150	22.1	292
	39600	22.2	281
LTE-TDD B41 @10Mhz,FRB	39700	22.07	251
	40620	22	244
	41540	21.93	247
LTE-FDD B71 @10Mhz,FRB	133172	21.5	478
	133297	21.96	436
	133422	21.7	443

Chapter 5 Interface Electrical Characteristics

5.1 Overview of This Chapter

- ✧ Working Storage Temperature
- ✧ Module IO Level
- ✧ Power Supply Voltage
- ✧ Electrostatic Characteristics
- ✧ Reliability Index

5.2 Working Storage Temperature

Table 5-1 Working Storage Temperature of YM310_A19 module

Parameters	Minimum Value	Maximum Value
Normal operating temperature	-30℃	75℃
Extreme working temperature	-40℃	85℃
Storage temperature	-40℃	90℃

5.3 Module IO Level

YM310 A19 The Module IO levels are as follows:

Table 5-2 Electrical Characteristics of YM310 A19 Module

Parameters	Parameter Description	Minimum Value	Maximum Value
VIH	Input a logical high-level voltage	0.65* VDD_EXT	VDD_EXT+0.3V
VIL	Input a logical low-level voltage	-	0.35*VDD_EXT
VOH	Output a logical high-level voltage	VDD_EXT-0.45V	VDD_EXT
VOL	Output a logical low-level voltage	0	0.45V

5.4 Power Supply Characteristics

The input power supply requirements for the YM310_A19 module are as follows:

Table 5-3 Working Voltage of YM310_A19 Module

Parameters	Minimum Value	Typical Value	Maximum Value
VBAT	3.3V	3.7V	4.2V

NOTE

✧ The power-on time of any interface of the module must not be earlier than the power-on time of the module; otherwise, it may cause abnormality or damage to the module.

5.5 Electrostatic Characteristics

During the internal design of the YM310 A19 module, corresponding ESD protection has been considered and implemented. However, ESD issues may also occur during the production assembly and experimental testing of the module. Therefore, application developers need to consider the ESD protection of the final product.

When customers design, in addition to referring to the recommended circuits in the document interface design, the following points also need to be noted:

✧ The PCB layout of protective devices should follow a "V" shaped line as much as possible and avoid a "T" shaped line.

✧ The ground plane around the module should be kept intact and not divided.

✧ During the production, assembly and laboratory testing of the module, attention should be paid to the ESD control of the surrounding environment and the operators.

Table 5-4 ESD Characteristics of YM310 A19

Test Port	Contact Discharge	Air Discharge	Unit
VBAT Power Supply	±4	±8	KV
Antenna Interface	±4	±8	KV
Other Interfaces	±0.5	±1	KV

5.6 Reliability Index

Table 5-5 YM310 A19 Reliability Test

Test Items	Test Conditions	Reference Standard	Test Result
Low-temperature work	Temperature: -40 ℃ Working mode: Normal operation Test duration: 24 hours	IEC60068-2-1	Visual inspection: Normal Function check: Normal
High-temperature work	Temperature: 85℃ Working mode: Normal operation Test duration: 24 hours	JESD22-A108-C	Visual inspection: Normal Function check: Normal
Temperature cycle	High temperature: 85℃ Low temperature: -40 ℃ Working mode: Normal operation Test duration: 30cycles; 1h+1h/cycle	JESD22-A105-B	Visual inspection: Normal Function check: Normal
Alternating Damp heat	High temperature: 55℃ Low temperature: 25℃ Humidity: 95%±3% Working mode: Normal operation Test duration: 6 cycles; 12h+12h/cycle	JESD22-A101-B	Visual inspection: Normal Function check: Normal Radio frequency index check: Normal
Temperature shock	High temperature: 85 ° C Low temperature: -40 ° C Temperature change time: <30 seconds Working mode: No packaging, no power supply, no startup Test duration: 100	JESD22-A106-B	Visual inspection: Normal Function check: Normal

	cycles; 15min+15min/cycle		
Drop test	Height 0.8m, 6 times each, dropped to a horizontal marble platform Working mode: no packaging, no power, no startup	IEC60068-2-32	Visual inspection: Normal Function check: Normal Radio frequency index check: Normal
Low-temperature storage	Temperature: -40 ° C Working mode: No packaging, no power supply, no startup Test duration: 24 hours	JESD22-A119-C	Visual inspection: Normal Function check: Normal
High-temperature storage	Temperature: 85 ° C Working mode: No packaging, no power supply, no startup Test duration: 24 hours	JESD22-A103-C	Visual inspection: Normal Function check: Normal

Chapter 6 Structure and Mechanical Characteristics

6.1 Overview of this Chapter

- ✧ Appearance
- ✧ Module Mechanical Dimensions

6.2 Appearance

The YM310 A19 module is a single-sided layout PCBA. The appearance diagram is as follows:



Figure 6-1 Appearance drawing of YM310 A19

6.3 Mechanical Dimensions

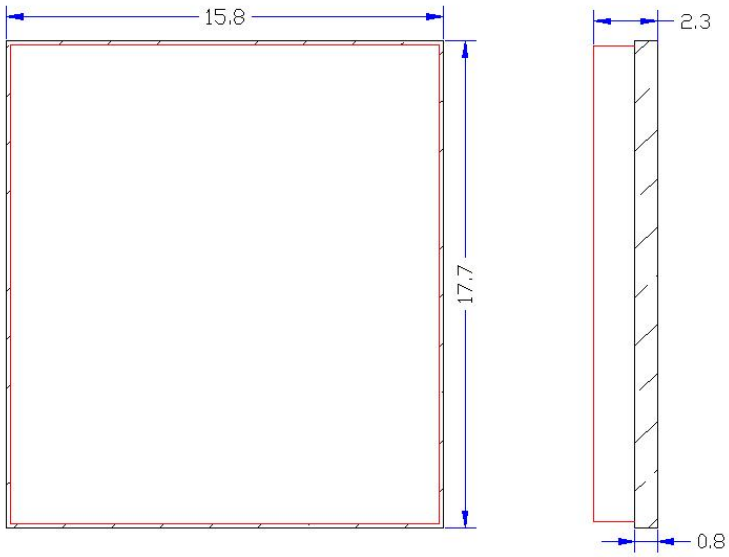


Figure 6-2 Front view and side view of the module (Unit: millimeters)

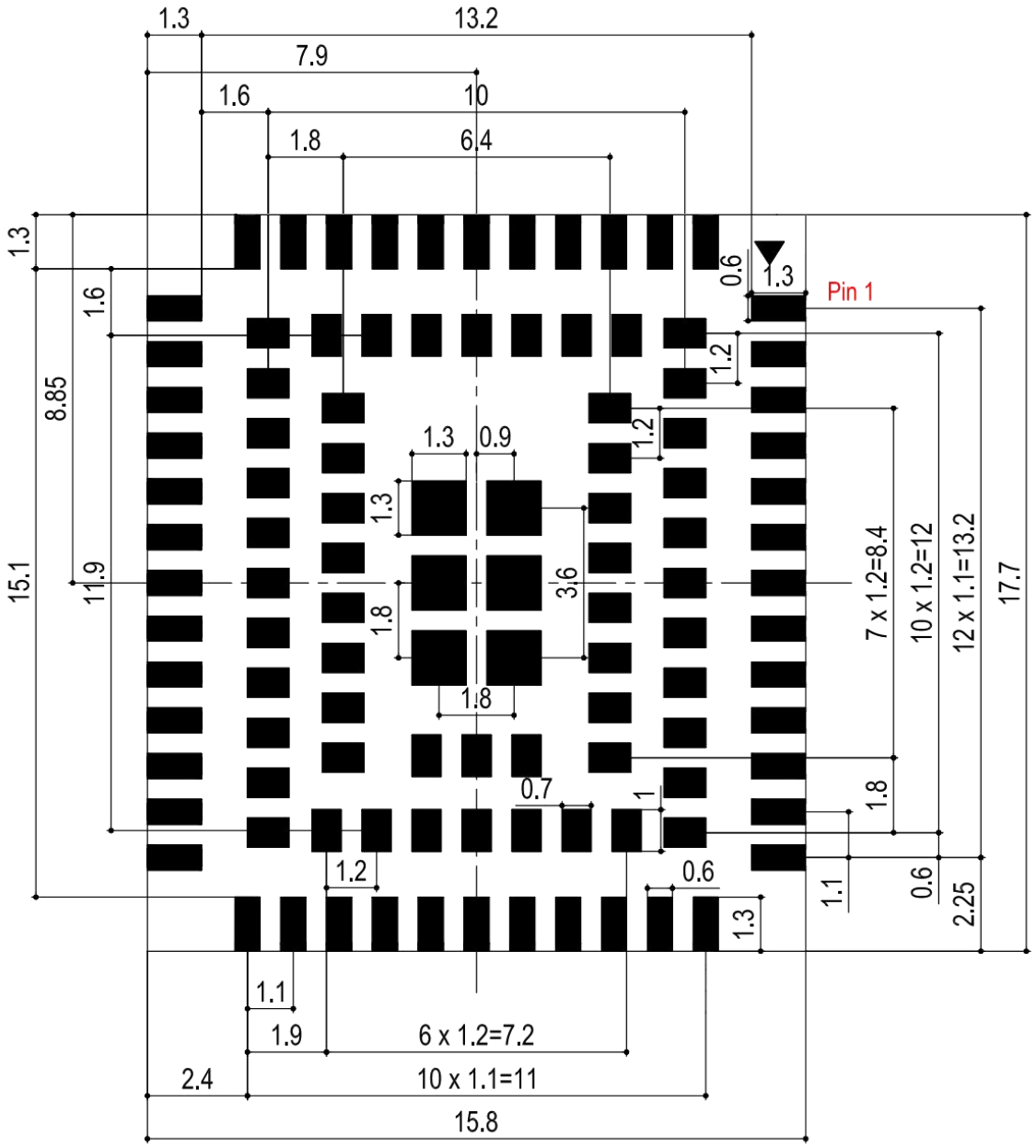


Figure 6-3 Module Bottom View (Unit: Millimeters)

Module recommended packaging:

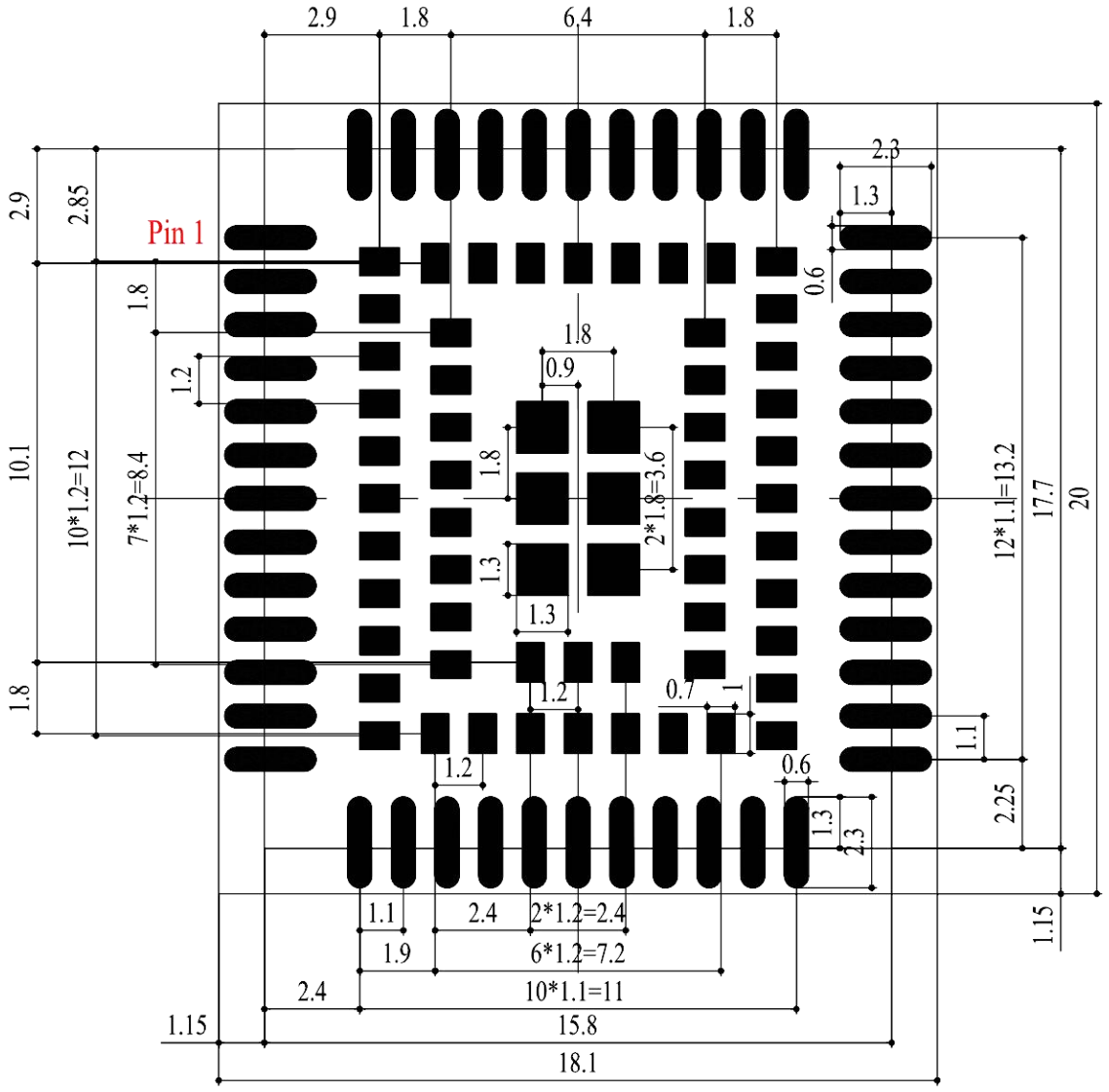


Figure 6-4 Module Recommended Packaging (Unit: Millimeters)

Chapter 7 Packaging and Production

7.1 Overview of this Chapter

- ✧ Module Packaging and Storage
- ✧ Production welding

7.2 Module Packaging and Storage

The YM310 A19 module is packaged with tape, with 1000PCS per plate, and each plate is shipped in a vacuum sealed bag.

The storage of YM310 A19 module needs to follow the following conditions:

- ✧ The moisture sensitivity level of the module is level 3.
- ✧ When the ambient temperature is less than 40 degrees Celsius and the air humidity is less than 90%, the module can be stored in a vacuum-sealed bag for 12 months.
- ✧ When the vacuum sealed bag is opened, if the ambient temperature of the module is below 30 degrees Celsius and the air humidity is less than 60%, the factory can complete the surface mount within 72 hours, and the module can directly undergo reflow soldering or other high-temperature processes.
- ✧ If the module is under other conditions, it needs to be baked before surface mount.
- ✧ If the module needs to be baked, after removing the module packaging, please bake it at 125 degrees Celsius (a fluctuation of up to 5 degrees Celsius is allowed) for 8 hours.

7.3 Production Welding

The YM310 A19 module uses tapered packaging, and the SMT line body needs to be equipped with a 32mm carrier.

- ✧ To ensure the quality of the module paste, the recommended thickness of the steel mesh corresponding to the pad part of the YM310 A19 module is 0.18mm.
- ✧ The recommended temperature for reflow soldering is 238-248 ° C and cannot exceed 248 ° C.
- ✧ When PCB double-sided layout is used, LCC module layout must be processed on the second side. To avoid module dropping, welding failure, and internal welding defects caused by module gravity induced flipping and reflow.

The recommended furnace temperature curve graph is shown as follows:

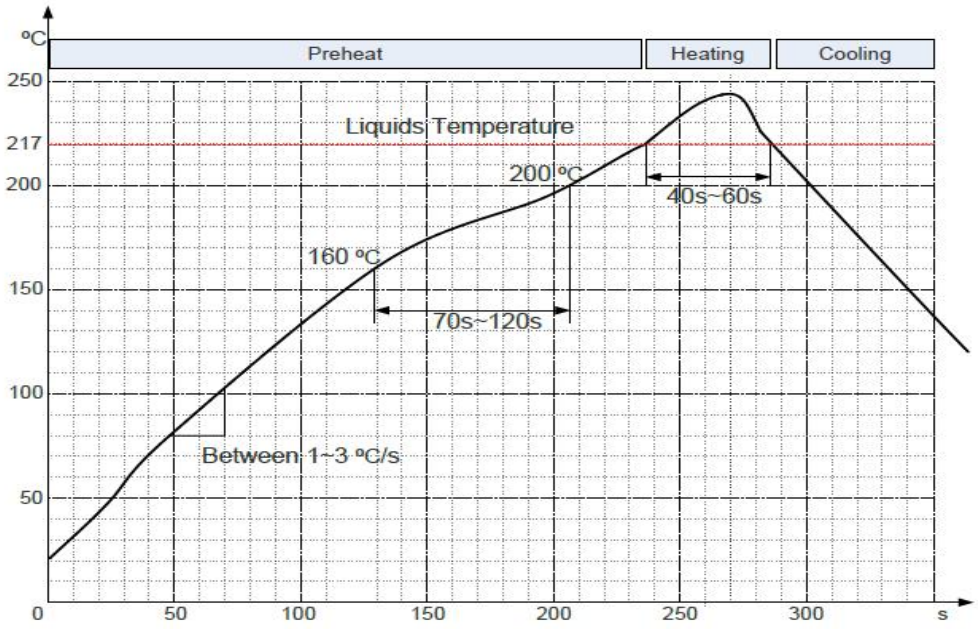


Figure 7-1 reflow soldering temperature curve graph

Table 7-1 Reflux Process Parameters Table

Temperature Zone	Time	Key Parameters
Preheating zone (40℃ - 165℃)		Heating rate: 1℃/s to 3℃/s
Uniform temperature zone (160℃ - 210℃)	(t1～t2)：70s～120s	
Reflux zone (>217℃)	(t3～t4)：40s～60s	Peak temperature: 238℃ to 248℃
Cooling Zone	Cooling rate: 2℃/s≤Slope≤5℃/s	

Chapter 8 Appendix

8.1 Overview of This Chapter

- ✧ Abbreviation
- ✧ Coding Method
- ✧ Usage safety and precautions

8.2 Abbreviation

Table 8-1 Abbreviations of Terms

Abbreviation	Full Name
3GPP	Third Generation Partnership Project
AP	Access Point
AMR	Adaptive Multi-rate
BER	Bit Error Rate
CCC	China Compulsory Certification
CDMA	Code Division Multiple Access
CE	European Conformity
CSD	Circuit Switched Data
CTS	Clear to Send
DC	Direct Current
DTR	Data Terminal Ready
DL	Down Link
DTE	Data Terminal Equipment
EU	European Union
EMC	Electromagnetic Compatibility
ESD	Electrostatic Discharge
HSDPA	High-Speed Downlink Packet Access
HSPA	Enhanced High Speed PacketAccess
HSUPA	High Speed Up-link Packet Access
IMEI	International Mobile Equipment Identity
LED	Light-Emitting Diode

LTE	Long Term Evolution
NC	Not Connected
PCB	Printed Circuit Board
PCM	Pulse Code Modulation
PDU	Protocol Data Unit
PMU	Power Management Unit
PPP	Point-to-point protocol
QPSK	Quadrature Phase Shift Keying
RF	Radio Frequency
RoHS	Restriction of the Use of CertainHazardous Substances
SMS	Short Message Service
TIS	Total Isotropic Sensitivity
TVS	Transient Voltage Suppressor
TX	Transmitting Direction
UART	Universal AsynchronousReceiver-Transmitter
UMTS	Universal Mobile Telecommunications System
USIM	Universal Subscriber Identity Module
USSD	Unstructured Supplementary Service Data
VSWR	Voltage Standing Wave Ratio
WCDMA	Wideband Code Division MultipleAccess
WWAN	Wireless Wide Area Network

8.3 Coding Method

Table 8-2 Time Slot Allocation Table for Different Levels of GPRS/EDGE

Slot class	DL slot number	UL slot number	Active slot number
1	1	1	2
2	2	1	3
3	2	2	3
4	3	1	4
5	2	2	4
6	3	2	4

7	3	3	4
8	4	1	5
9	3	2	5
10	4	2	5
11	4	3	5
12	4	4	5

Table 8-3 Maximum Rate of GPRS

GPRS coding scheme	Max data rata（4 slots）	Modulation type
CS 1 = 9.05 kb/s / time slot	36.2 kb/s	GMSK
CS 2 = 13.4 kb/s / time slot	53.6 kb/s	GMSK
CS 3 = 15.6 kb/s / time slot	62.4 kb/s	GMSK
CS 4 = 21.4 kb/s / time slot	85.6 kb/s	GMSK

Table 8-4 Maximum Rate of EDGE

GPRS coding scheme	Max data rata（4 slots）	Modulation Type
MCS 1 = 8.8 kb/s/ time slot	35.2 kb/s	GMSK
MCS 2 = 11.2 kb/s/ time slot	44.8 kb/s	GMSK
MCS 3 = 14.8 kb/s/ time slot	59.2 kb/s	GMSK
MCS 4 = 17.6 kb/s/ time slot	70.4 kb/s	GMSK
MCS 5 = 22.4 kb/s/ time slot	89.6 kb/s	8PSK
MCS 6 = 29.6 kb/s/ time slot	118.4 kb/s	8PSK
MCS 7 = 44.8 kb/s/ time slot	179.2 kb/s	8PSK
MCS 8 = 54.4 kb/s/ time slot	217.6 kb/s	8PSK
MCS 9 = 59.2 kb/s/ time slot	236.8 kb/s	8PSK

Table 8-5 Maximum Rate of LTE-FDD DL

LTE-FDD device category	Max data rate(peak)	Modulation Type
Category 1	10Mbps	QPSK/16QAM/64QAM
Category 2	50Mbps	QPSK/16QAM/64QAM
Category 3	100Mbps	QPSK/16QAM/64QAM
Category 4	150Mbps	QPSK/16QAM/64QAM

Table 8-6 Maximum UL Rate of LTE-FDD

LTE-FDD device category	Max data rate(peak)	Modulation type
Category 1	5Mbps	QPSK/16QAM
Category 2	25Mbps	QPSK/16QAM
Category 3	50Mbps	QPSK/16QAM
Category 4	50Mbps	QPSK/16QAM

8.4 Usage Safety and Precautions

For the safe use of wireless devices, please inform the user of relevant security information through the terminal device:

- ✧ Interference: When the use of wireless devices is prohibited or the use of such devices may cause interference and safety issues with electronic devices, please turn off the wireless devices. Because the terminal will send and receive radio frequency signals when it is turned on. When close to televisions, radios, computers or other electrical devices, it will cause interference to them.
- ✧ Medical equipment: In medical and health care facilities where the use of wireless devices is explicitly prohibited, please follow the regulations of such facilities and turn off this device. Some wireless devices may interfere with medical equipment, causing it to malfunction or resulting in errors. If interference occurs, please turn off the wireless device and consult a doctor.
- ✧ Flammable and explosive areas: In flammable and explosive areas, please turn off your wireless devices and follow the relevant signs and instructions to avoid explosion or fire. Such as: gas stations, fuel areas, chemical product areas, as well as chemical transportation and storage facilities, areas with explosion hazard signs, areas with "Turn off radio equipment" signs, etc.
- ✧ Traffic safety: Please abide by the local laws or regulations of your country or region regarding the use of wireless devices while driving.
- ✧ Aviation safety: When taking a flight, please abide by the relevant regulations and rules of the airline regarding the use of wireless devices. Before takeoff, please turn off the wireless devices to avoid wireless signals interfering with the aircraft control signals.
- ✧ Environmental protection: Please comply with local laws and regulations regarding the handling of equipment packaging materials, equipment or its accessories, and support recycling actions.
- ✧ Emergency call: This device uses wireless signals for transmission. Therefore, it cannot be guaranteed that the network will be connected in all circumstances. Thus, in an emergency, this wireless device cannot be used as the sole means of contact.

FCC MODULAR APPROVAL INFORMATION EXAMPLES for Manual

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:

(1) This device may not cause harmful interference.

(2) This device must accept any interference received, including interference that may cause undesired operation.

CAUTION: Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

NOTE: This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

FCC Radiation Exposure Statement: This equipment complies with FCC radiation exposure limits set forth for an uncontrolled environment. This equipment should be installed and operated with minimum distance 20cm between the radiator & your body.

OEM INTEGRATION INSTRUCTIONS:

This device is intended only for OEM integrators under the following conditions:

The module must be installed in the host equipment such that 20 cm is maintained between the antenna and users, and the transmitter module may not be co-located with any other transmitter or antenna. The module shall be only used with the internal on-board antenna that has been originally tested and certified with this module. External antennas are not supported. As long as these 3 conditions above are met, further transmitter test will not be required.

However, the OEM integrator is still responsible for testing their end-product for any additional compliance requirements required with this module installed (for example, digital device emissions, PC peripheral requirements, etc.). The end-product may need Verification testing, Declaration of Conformity testing, a Permissive Class II Change or new Certification. Please involve a FCC certification specialist in order to determine what will be exactly applicable for the end-product.

Validity of using the module certification:

In the event that these conditions cannot be met (for example certain laptop configurations or co-location with another transmitter), then the FCC authorization for this module in combination with the host equipment is no longer considered valid and the FCC ID of the module cannot be used on the final product. In these circumstances, the OEM integrator will be responsible for re-evaluating the end product (including the transmitter) and obtaining a separate FCC authorization. In such cases, please involve a FCC certification specialist in order to determine if a Permissive Class II Change or new Certification is required.

Upgrade Firmware:

The software provided for firmware upgrade will not be capable to affect any RF parameters as certified for the FCC for this module, in order to prevent compliance issues.

End product labeling:

This transmitter module is authorized only for use in device where the antenna may be installed such that 20 cm may be maintained between the antenna and users. The final end product must be labeled in a visible area with the following: “ Contains FCC ID: 2BG5S-YM310-A19 ”.

Information that must be placed in the end user manual:

The OEM integrator has to be aware not to provide information to the end user regarding how to install or remove this RF module in the user's manual of the end product which integrates this module. The end user manual shall include all required regulatory information/warning as show in this manual.

“ CAUTION : Exposure to Radio Frequency Radiation.
Antenna shall be mounted in such a manner to minimize the potential for human contact during normal operation. The antenna should not be contacted during operation to avoid the possibility of exceeding the FCC radio frequency exposure limit.

2.2 List of applicable FCC rules

List the FCC rules that are applicable to the modular transmitter. These are the rules that specifically establish the bands of operation, the power, spurious emissions, and operating fundamental frequencies. DO NOT list compliance to unintentional-radiator rules (Part 15 Subpart B) since that is not a condition of a module grant that is extended to a host manufacturer. See also Section 2.10 below concerning the need to notify host manufacturers that further testing is required.3

Explanation: This module meets the requirements of FCC part 15C(15.247). part 15E(15.407)

2.3 Summarize the specific operational use conditions

Describe use conditions that are applicable to the modular transmitter, including for example any limits on antennas, etc. For example, if point-to-point antennas are used that require reduction in power or compensation for cable loss, then this information must be in the instructions. If the use condition limitations extend to professional users, then instructions must state that this information also extends to the host manufacturer ’s instruction manual. In addition, certain information may also be needed, such as peak gain per frequency band and minimum gain, specifically for master devices in 5 GHz DFS bands.

Explanation: The EUT has a Chip Antenna, and the antenna use a permanently attached antenna which is not replaceable.

2.4 Limited module procedures

If a modular transmitter is approved as a “ limited module, ” then the module manufacturer is responsible for approving the host environment that the limited module is used with. The manufacturer of a limited module must describe, both in the filing and in the installation instructions, the alternative means that the limited module manufacturer uses to verify that the host meets the necessary requirements to satisfy the module limiting conditions.

A limited module manufacturer has the flexibility to define its alternative method to address the conditions that limit the initial approval, such as: shielding, minimum signaling amplitude, buffered modulation/data inputs, or power supply regulation. The alternative method could include that the limited module manufacturer reviews detailed test data or host designs prior to giving the host manufacturer approval.

This limited module procedure is also applicable for RF exposure evaluation when it is necessary to demonstrate compliance in a specific host. The module manufacturer must state how control of the product into which the modular transmitter will be installed will be maintained such that full compliance of the product is always ensured. For additional hosts other than the specific host originally granted with a limited module, a Class II permissive change is required on the module grant to register the additional host as a specific host also approved with the module.

Explanation: The module is not a limited module.

2.5 Trace antenna designs

For a modular transmitter with trace antenna designs, see the guidance in Question 11 of KDB Publication 996369 D02 FAQ – Modules for Micro-Strip Antennas and traces. The integration information shall include for the TCB review the integration instructions for the following aspects:

layout of trace design, parts list (BOM), antenna, connectors, and isolation requirements.

a) Information that includes permitted variances (e.g., trace boundary limits, thickness, length, width, shape(s), dielectric constant, and impedance as applicable for each type of antenna);

b) Each design shall be considered a different type (e.g., antenna length in multiple(s) of frequency, the wavelength, and antenna shape (traces in phase) can affect antenna gain and must be considered);

c) The parameters shall be provided in a manner permitting host manufacturers to design the printed circuit (PC) board layout;

d) Appropriate parts by manufacturer and specifications;

e) Test procedures for design verification; and

f) Production test procedures for ensuring compliance.

The module grantee shall provide a notice that any deviation(s) from the defined parameters of the antenna trace, as described by the instructions, require that the host product manufacturer must notify the module grantee that they wish to change the antenna trace design. In this case, a Class II permissive change application is required to be filed by the grantee, or the host manufacturer can take responsibility through the change in FCC ID (new application) procedure followed by a Class II permissive change application.

Explanation: Yes, The module with trace antenna designs, and This manual has been shown the layout of trace design, antenna, connectors, and isolation requirements.

2.6 RF exposure considerations

It is essential for module grantees to clearly and explicitly state the RF exposure conditions that permit a host product manufacturer to use the module. Two types of instructions are required for RF exposure information: (1) to the host product manufacturer, to define the application conditions (mobile, portable – xx cm from a person's body); and (2) additional text needed for the host product manufacturer to provide to end users in their end-product manuals. If RF exposure statements and use conditions are not provided, then the host product manufacturer is required to take responsibility of the module through a change in FCC ID (new application).

Explanation: This module complies with FCC RF radiation exposure limits set forth for an uncontrolled environment, This equipment should be installed and operated with a minimum distance of 20 centimeters between the radiator and your body."This module is designed to comply with the FCC statement, FCC ID is: 2BG5S-YM310-A19.

2.7 Antennas

A list of antennas included in the application for certification must be provided in the instructions. For modular transmitters approved as limited modules, all applicable professional installer instructions must be included as part of the information to the host product manufacturer. The antenna list shall also identify the antenna types (monopole, PIFA, dipole, etc. (note that for example an "omni-directional antenna" is not considered to be a specific "antenna type")).

For situations where the host product manufacturer is responsible for an external connector, for example with an RF pin and antenna trace design, the integration instructions shall inform the installer that unique antenna connector must be used on the Part 15 authorized transmitters used in the host product. The module manufacturers shall provide a list of acceptable unique connectors.

Explanation: The EUT has a Chip Antenna, and the antenna use a permanently attached antenna which is unique.

2.8 Label and compliance information

Grantees are responsible for the continued compliance of their modules to the FCC rules. This includes advising host product manufacturers that they need to provide a physical or e-label stating "Contains FCC ID" with their finished product. See Guidelines for Labeling and User Information for RF Devices – KDB Publication 784748.

Explanation: The host system using this module, should have label in a visible area indicated the following texts: "Contains FCC ID: 2BG5S-YM310-A19"

2.9 Information on test modes and additional testing requirements5

Additional guidance for testing host products is given in KDB Publication 996369 D04 Module Integration Guide. Test modes should take into consideration different operational conditions for a stand-alone modular transmitter in a host, as well as for multiple simultaneously transmitting modules or other transmitters in a host product.

The grantee should provide information on how to configure test modes for host product evaluation for different operational conditions for a stand-alone modular transmitter in a host, versus with multiple, simultaneously transmitting modules or other transmitters in a host.

Grantees can increase the utility of their modular transmitters by providing special means, modes, or instructions that simulates or characterizes a connection by enabling a transmitter. This can greatly simplify a host manufacturer's determination that a module as installed in a host complies with FCC requirements.

Explanation: Top band can increase the utility of our modular transmitters by providing instructions that simulates or characterizes a connection by enabling a transmitter.

2.10 Additional testing, Part 15 Subpart B disclaimer

The grantee should include a statement that the modular transmitter is only FCC authorized for the specific rule parts (i.e., FCC transmitter rules) listed on the grant, and that the host product manufacturer is responsible for compliance to any other FCC rules that apply to the host not covered by the modular transmitter grant of certification. If the grantee markets their product as being Part 15 Subpart B compliant (when it also contains unintentional-radiator digital circuity), then the grantee shall provide a notice stating that the final host product still requires Part 15 Subpart B compliance testing with the modular transmitter installed.

Explanation: The module without unintentional-radiator digital circuity, so the module does not require an evaluation by FCC Part 15 Subpart B. The host shoule be evaluated by the FCC Subpart B.

2.11 Note EMI Considerations

Note that a host manufacture is recommended to use D04 Module Integration Guide recommending as "best practice" RF design engineering testing and evaluation in case non-linear interactions generate additional non-compliant limits due to module placement to host components or properties

For standalone mode, reference the guidance in D04 Module Integration Guide and for simultaneous mode7; see D02 Module Q&A Question 12, which permits the host manufacturer to confirm compliance.
Explanation: The module without unintentional-radiator digital circuity, so the module does not require an evaluation by FCC Part 15 Subpart B. The host shoule be evaluated by the FCC Subpart B.

2.12 How to make changes

Since . only Grantees are permitted to make permissive changes, it is recommended that module manufactures provide contact information and some guidance to host providers in the integration instructions if they expect their module will be used differently than granted.
Explanation: "Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment."